Double Advanced

Daily math workbook

Ten weeks. One day at a time.

Find your day

Each day has one continuous section. Saturdays and Sundays go with the slides. Monday through Friday are short practice sessions.

During a lesson

P labels the lesson problems. F is your turn after an example. D is a discussion. E is the exit ticket. Match the slide label to the same label on paper.

During the week

Built-in practice is already part of the day. It replaces other questions. It is not an extra assignment. Work for 15-20 minutes, then tell your tutor what remains.

Reference pages

Study guides and paper tools sit beside the days that use them. Read or use these pages when needed. They are not extra questions to finish.

Answers

The tutor has a separate guide for each week. Review the answers after you have tried the questions.

Find a day

Each cell gives the day number and the pages to print.

Week	Sat	Sun	Mon	Tue	Wed	Thu	Fri
1	Day 001 p. 3-14	Day 002 p. 15-25	Day 003 p. 26-27	Day 004 p. 28-29	Day 005 p. 30-31	Day 006 p. 32-33	Day 007 p. 34-35
2	Day 008 p. 36-42	Day 009 p. 43-52	Day 010 p. 53-54	Day 011 p. 55-56	Day 012 p. 57-58	Day 013 p. 59-60	Day 014 p. 61-62
3	Day 015 p. 63-67	Day 016 p. 68-78	Day 017 p. 79-80	Day 018 p. 81-82	Day 019 p. 83-84	Day 020 p. 85-86	Day 021 p. 87-88
4	Day 022 p. 89-95	Day 023 p. 96-105	Day 024 p. 106-107	Day 025 p. 108-109	Day 026 p. 110-111	Day 027 p. 112-113	Day 028 p. 114-115
5	Day 029 p. 116-120	Day 030 p. 121-130	Day 031 p. 131-132	Day 032 p. 133-134	Day 033 p. 135-136	Day 034 p. 137-138	Day 035 p. 139-140
6	Day 036 p. 141-145	Day 037 p. 146-155	Day 038 p. 156-157	Day 039 p. 158-159	Day 040 p. 160-161	Day 041 p. 162-163	Day 042 p. 164-165
7	Day 043 p. 166-171	Day 044 p. 172-184	Day 045 p. 185-186	Day 046 p. 187-188	Day 047 p. 189-190	Day 048 p. 191-192	Day 049 p. 193-194
8	Day 050 p. 195-201	Day 051 p. 202-212	Day 052 p. 213-214	Day 053 p. 215-216	Day 054 p. 217-218	Day 055 p. 219-220	Day 056 p. 221-222
9	Day 057 p. 223-227	Day 058 p. 228-240	Day 059 p. 241-242	Day 060 p. 243-244	Day 061 p. 245-246	Day 062 p. 247-248	Day 063 p. 249-250
10	Day 064 p. 251-256	Day 065 p. 257-267	Day 066 p. 268-269	Day 067 p. 270-271	Day 068 p. 272-273	Day 069 p. 274-275	Day 070 p. 276-277

Starting line

This shows what to practice next. Skip anything you have not learned.

Fact snapshot (up to 2 minutes)

1. 5 × 4	2. 3 × 6	3. 7 × 11
4. 7 × 8	5. 8 × 8	6. 4 × 3
7. 7 × 6	8. 5 × 7	9. 8 × 11
10. 2 × 11	11. 7 × 9	12. 6 × 9
13. 8 × 2	14. 12 × 10	15. 4 × 5
16. 7 × 4	17. 3 × 5	18. 2 × 4
19. 4 × 12	20. 6 × 10	21. 11 × 2
22. 2 × 5	23. 6 × 11	24. 6 × 3
25. 8 × 7	26. 6 × 12	27. 7 × 10
28. 8 × 6	29. 7 × 2	30. 12 × 7

Stop when the tutor calls time. Accuracy matters. Compare only with your own future work.

Part A. Computation

1. 426 × 23

Answer: _____

2. 1,368 ÷ 6

Answer: _____

3. 2,184 ÷ 24

Answer: _____

4. 4.65 + 13.8

Answer: _____

5. 8.1 − 3.47

Answer: _____

6. 0.7 × 0.4

Answer: _____

Part A. Computation

7. $6.3 \div 0.9$

Answer: _____

8. $\frac{3}{5} \times \frac{5}{6}$

Answer: _____

9. $\frac{5}{6} + \frac{1}{4}$

Answer: _____

10. How many $\frac{1}{4}$s fit in 3?

Answer: _____

11. $\frac{1}{5} \div 3$

Answer: _____

12. $2\ \frac{1}{3} \times 3$

Answer: _____

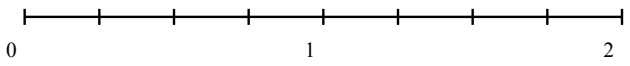
Part A. Computation

After item 13, take a 5-minute break. Part B starts with item 14.

13. Compare $\frac{4}{9}$ and $\frac{1}{2}$.

Answer: _____

14. Place $\frac{3}{4}$, $\frac{4}{3}$, $\frac{5}{8}$, and $1\ \frac{1}{2}$ on the line.



Answer: _____

15. Order 0.6, 0.56, and 0.506 from least to greatest.

Answer: _____

16. A bag has 3 red tiles for every 2 blue. There are 15 red tiles. How many blue tiles?

Answer: _____

17. Find 25% of 64.

Answer: _____

18. Evaluate 4n + 3 when n = 6.

Answer: _____

Part B. Reasoning

19. Morning temperature is −4°C. Noon is 2°C. How much warmer is noon?

Answer: _____

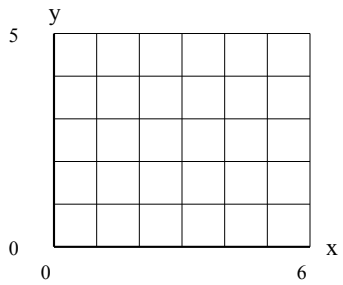
20. A rectangle is $\frac{3}{4}$ m long and $\frac{2}{3}$ m wide. Find its area.

Answer: _____

21. A box is 4 cm by 3 cm by 5 cm. Find its volume.

Answer: _____

22. Plot A(2, 4) and B(5, 4). How far apart are they?



23. Explain why $\frac{2}{3} \times 9$ is smaller than 9.

Answer: _____

24. Persistence problem. A pattern starts 2, 5, 9, 14. Each jump grows by 1. Find the next two terms.

Answer: _____

Part B. Reasoning

25. Persistence problem. Three boxes hold 18 tiles. Each box has two more than the box before it. Find the three counts.

Answer: _____

Fractions on one number line

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Compare $\frac{3}{7}$ and $\frac{5}{7}$. Write <, >, or =.

Answer: _____

P2. Compare $\frac{4}{5}$ and $\frac{4}{9}$. Use strips of the same length.

Answer: _____

P3. Is $\frac{5}{12}$ less than or greater than $\frac{1}{2}$?

Answer: _____

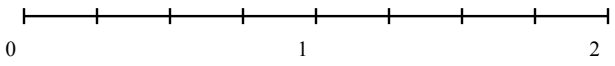
Fractions on one number line

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Compare $\frac{3}{5}$ and $\frac{7}{12}$. Rename both using sixtieths.

Answer: _____

P5. On a 0-to-2 line, place $\frac{5}{4}$, $1\ \frac{3}{4}$, and 0.5.



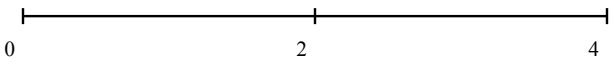
P6. Compare $\frac{7}{10}$ and 0.68.

Answer: _____

Fractions on one number line

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. A line has ticks at 0, 2, and 4. Place 2 $\frac{1}{2}$.



P8. Three identical muffins are shared equally by four friends. How much muffin does each friend receive?

Answer: _____

F. Complete: $\frac{5}{6}$ = __ twelfths. Compare $\frac{5}{6}$ and $\frac{3}{4}$.

Answer: _____

Fractions on one number line

Follow the slide labels. P = problem; F = your turn; D = discussion.

D. Nora says $\frac{4}{9}$ is greater than $\frac{4}{5}$ because 9 is greater than 5. Explain and repair her claim.

Answer: _____

S. To compare fractions, I first keep ___ the same. Then I can ___.

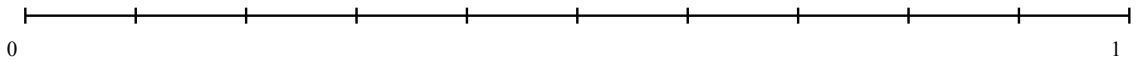
Answer: _____

Exit ticket

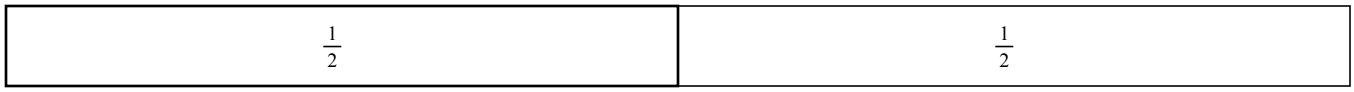
Work on your own. Show a model or reason when the question asks for one.

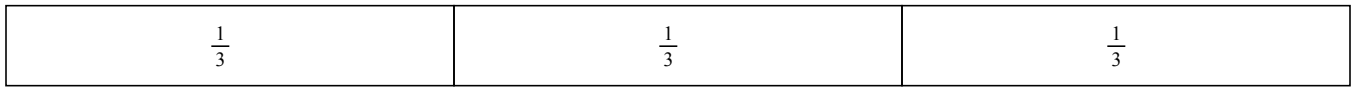
E. Order $\frac{3}{4}$, $\frac{7}{10}$, and 0.72 from least to greatest. Place them on a 0-to-1 line.

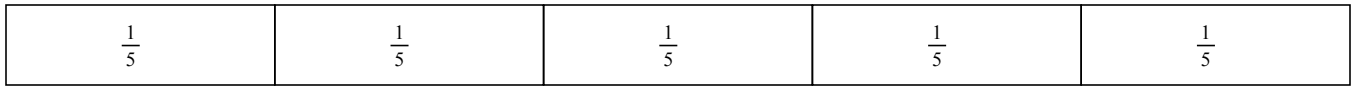
Answer: _____

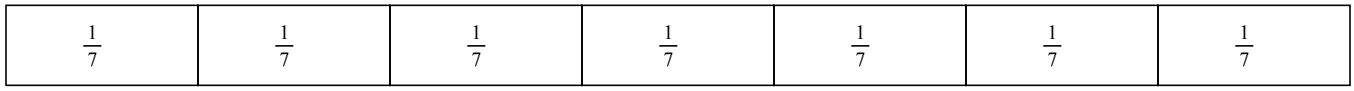


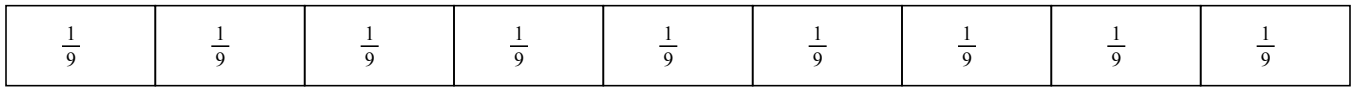
Fraction strips: equal wholes

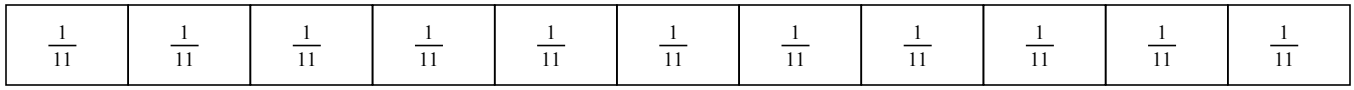












Recall

Work from memory, then check with your tutor.

R1. 7×8

Answer: _____

R2. $96 \div 12$

Answer: _____

R3. $\frac{1}{2} + \frac{1}{4}$

Answer: _____

R4. $0.6 + 0.25$

Answer: _____

S. Return here near the end of the lesson

When I divide, I first ask whether I need the number of ___ or the size of ___.

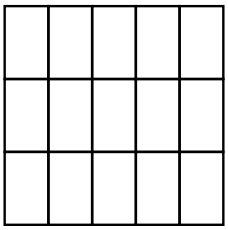
Fraction operations with models

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Five bags each hold $\frac{3}{4}$ cup of seeds. How many cups altogether?

Answer: _____

P2. A rectangle is $\frac{3}{5}$ meter long and $\frac{2}{3}$ meter wide. Find its area using equal parts.



P3. Find $\frac{2}{3}$ of $1\ \frac{1}{2}$ cups.

Answer: _____

Fraction operations with models

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Without multiplying first, compare $\frac{3}{4} \times 8$ and $\frac{5}{4} \times 8$ with 8. Then compute.

Answer: _____

P5. Three meters of ribbon are cut into $\frac{1}{4}$-meter pieces. How many pieces?

Answer: _____

P6. A $\frac{1}{4}$-liter juice sample is shared equally among three cups. How much goes in each cup?

Answer: _____

Fraction operations with models

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. Write one story for $2 \div \frac{1}{5}$ and one for $\frac{1}{5} \div 2$. Solve both.

Answer: _____

F. A model has 4 columns and 3 rows. Shade $\frac{3}{4}$, then $\frac{2}{3}$ of it. Complete __ × __ = __.

Answer: _____

D. Eli says $3 \div \frac{1}{4} = \frac{3}{4}$, because division makes numbers smaller. Show why this fails.

Answer: _____

Mixed review

M1. 13×4

Answer: _____

M2. $72 \div 9$

Answer: _____

M3. $\frac{1}{3} + \frac{1}{6}$

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. A board is 5 meters long. Each shelf needs $\frac{1}{2}$ meter. How many shelves can be cut?

Answer: _____

Fractions and equal parts

The big idea

The whole stays the same when fractions are compared. Multiplication can take part of an amount. Division can count fractional groups or split a fraction into shares.

Words to know

Numerator: The top number counts equal parts. Example: In $\frac{3}{5}$, count three fifths.

Denominator: The bottom number names how many equal parts make one whole. Example: Five fifths make one whole.

Equivalent fractions: Different names for the same amount. Example: $\frac{1}{2} = \frac{3}{6}$.

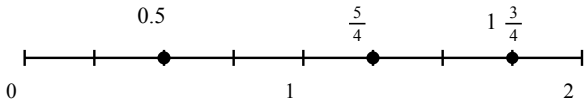
Quotient: The result of division. Example: $3 \div 4 = \frac{3}{4}$.

Scaling: Multiplying an amount to change its size. Example: $\frac{1}{2} \times 8$ takes half of eight.

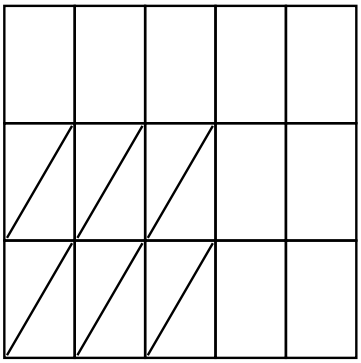
Unit fraction: A fraction with numerator 1. Example: $\frac{1}{5}$ is one fifth.

Product: The result of multiplication. Example: $\frac{1}{2} \times 8$ has product 4.

How it works



One whole keeps the same length. A fraction names equal parts of one whole. The size of that whole stays fixed.



$\frac{2}{3}$ of $\frac{3}{5}$ is $\frac{6}{15}$, or $\frac{2}{5}$. Multiplication can take part of an amount. Division can count groups or find the size of each share.

Worked examples

1. Compare $\frac{2}{3}$ and $\frac{3}{4}$.

Use one whole for both fractions.

Split the whole into 12 equal parts.

$\frac{2}{3} = \frac{8}{12}$; $\frac{3}{4} = \frac{9}{12}$.

Eight twelfths is less than nine twelfths.

Answer: $\frac{2}{3} < \frac{3}{4}$

2. Find $\frac{3}{4}$ of $\frac{2}{3}$ cup.

Split a whole into 3 equal columns and 4 equal rows.

Select 2 columns, then 3 fourths of those columns.

The overlap is 6 of 12 pieces: $\frac{6}{12} = \frac{1}{2}$.

Answer: $\frac{1}{2}$ cup

Common mistakes

Places $\frac{5}{4}$ between 0 and 1. Why: Thinks all fractions are less than one. Fix: Build four fourths, then add one fourth.

Orders by denominator size. Why: Confuses number of parts with part size. Fix: Compare equal-length strips cut into fifths and ninths.

Says multiplication always increases an amount. Why: Uses only whole-number experience. Fix: Take half of a familiar amount.

Try it

1. Compare $\frac{2}{9}$ and $\frac{7}{9}$.

2. Which is closer to 1: $\frac{7}{8}$ or $\frac{4}{5}$?

3. Find $\frac{3}{4} \times \frac{4}{5}$.

4. Find $\frac{3}{5}$ of 1 $\frac{2}{3}$ cups.

5. Is $\frac{2}{3} \times 9$ smaller or larger than 9? Compute to check.

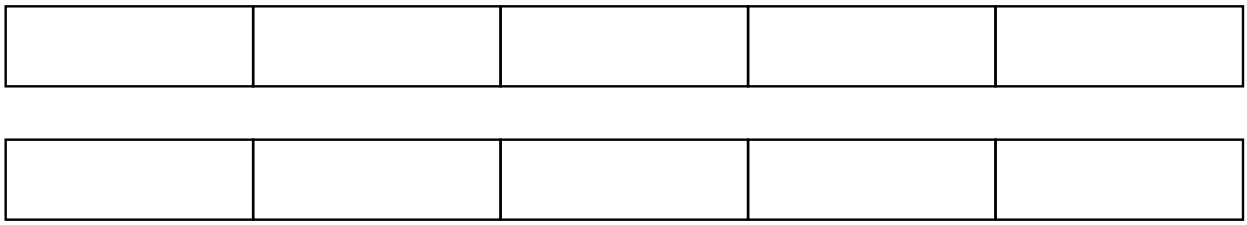
Answers

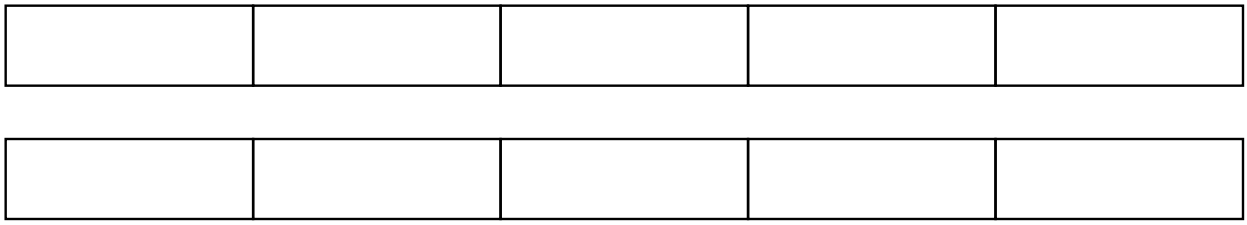
1. $\frac{2}{9} < \frac{7}{9}$ 2. $\frac{7}{8}$ 3. $\frac{3}{5}$ 4. 1 cup 5. Smaller; 6

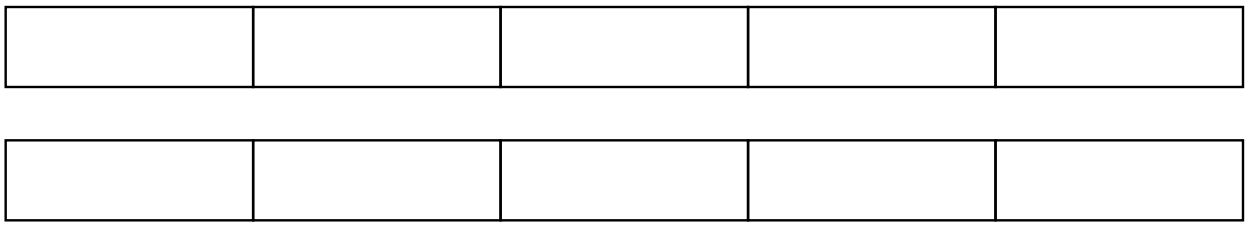
Where this connects

This builds on equal sharing and multiplication facts. Next, ratios compare amounts that change together.

Tape Diagram Template







Daily practice

A. Warm-up (3 minutes)

1. 3×3

Answer: _____

2. 3×7

Answer: _____

3. 3×4

Answer: _____

4. 3×9

Answer: _____

B. Practice

5. $24 \div 2$

Answer: _____

6. 8×5

Answer: _____

7. 4×4

Answer: _____

8. Compare $\frac{1}{3}$ and $\frac{1}{4}$.

Answer: _____

Practice and thinking

Built-in practice: Fractions: compare and combine

9. Compare $\frac{2}{9}$ and $\frac{7}{9}$.

Answer: _____

10. Compare $\frac{2}{4}$ and $\frac{2}{5}$.

Answer: _____

11. Order $\frac{2}{5}$, 0.45, and $\frac{1}{2}$ from least to greatest.

Answer: _____

12. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

Answer: _____

C. Think

13. Nora says $\frac{4}{9}$ is greater than $\frac{4}{5}$ because 9 is greater than 5. Explain and repair her claim.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Write $\frac{1}{2}$ as a decimal and a percent.

Answer: _____

2. Write $\frac{1}{4}$ as a decimal and a percent.

Answer: _____

3. Write $\frac{3}{4}$ as a decimal and a percent.

Answer: _____

4. Write $\frac{1}{8}$ as a decimal and a percent.

Answer: _____

B. Practice

5. 3×5

Answer: _____

6. 2×6

Answer: _____

7. $60 \div 5$

Answer: _____

8. Find both $4 \div \frac{1}{5}$ and $\frac{1}{5} \div 4$.

Answer: _____

Practice and thinking

Built-in practice: Fraction operations

9. Find $3 \times \frac{2}{3}$.

Answer: _____

10. $\frac{2}{3} \times \frac{2}{5}$

Answer: _____

11. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

Answer: _____

12. Share $\frac{1}{3}$ liter equally among 2 cups. How much per cup?

Answer: _____

C. Think

13. Eli says $3 \div \frac{1}{4} = \frac{3}{4}$, because division makes numbers smaller. Show why this fails.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. 3.2×10

Answer: _____

2. $300 \div 100$

Answer: _____

3. Double 3.5

Answer: _____

4. Half of 18

Answer: _____

B. Practice

5. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

Answer: _____

6. Write a story for $\frac{1}{6} \div 3$ and solve it.

Answer: _____

7. 4×6

Answer: _____

8. $\frac{1}{6} + \frac{1}{6}$

Answer: _____

Practice and thinking

B. Practice, continued

9. 300 ÷ 100

Answer: _____

10. 3 × 2

Answer: _____

11. 6 × 6

Answer: _____

12. A box is 6 cm by 6 cm by 3 cm. Find its volume.

Answer: _____

C. Think: weekly puzzle

13. A card shows a fraction. Its denominator is 8. It lies between $\frac{1}{2}$ and $\frac{3}{4}$. What fraction is on the card?

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Round 348 to the nearest hundred.

Answer: _____

2. Round 3.72 to the nearest whole number.

Answer: _____

3. Use 20×5 to estimate 19.8×5.1.

Answer: _____

4. Use $60 \div 3$ to estimate $59.4 \div 3.1$.

Answer: _____

B. Practice

5. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

Answer: _____

6. Write a story for $\frac{1}{6} \div 3$ and solve it.

Answer: _____

7. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

Answer: _____

8. A box is 5 cm by 2 cm by 3 cm. Find its volume.

Answer: _____

Practice and thinking

Built-in practice: Multiplication and division facts

9. 6×8

Answer: _____

10. $48 \div 6$

Answer: _____

11. 7×9

Answer: _____

12. $63 \div 7$

Answer: _____

C. Think

13. Is $\frac{2}{3} \times 9$ smaller or larger than 9? Compute to check.

Answer: _____

14. Eli says $3 \div \frac{1}{4} = \frac{3}{4}$, because division makes numbers smaller. Show why this fails.

Answer: _____

Weekly check

20 minutes. No calculator. Show your work.

1. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

Answer: _____

2. Write a story for $\frac{1}{6} \div 3$ and solve it.

Answer: _____

3. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

Answer: _____

4. Find $\frac{1}{2} + \frac{1}{3}$.

Answer: _____

5. A box is 6 cm by 3 cm by 3 cm. Find its volume.

Answer: _____

Weekly check

6. $72 \div 6$

Answer: _____

7. $6.5 + 0.25$

Answer: _____

8. $500 \div 100$

Answer: _____

9. 5×4

Answer: _____

10. Nora says $\frac{4}{9}$ is greater than $\frac{4}{5}$ because 9 is greater than 5. Explain and repair her claim.

Answer: _____

Recall

Work from memory, then check with your tutor.

R1. Share $\frac{1}{3}$ liter equally among 2 cups. How much per cup?

Answer: _____

R2. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

Answer: _____

R3. Compare $\frac{1}{3}$ and $\frac{1}{4}$.

Answer: _____

R4. Find $\frac{3}{4} \times \frac{4}{5}$.

Answer: _____

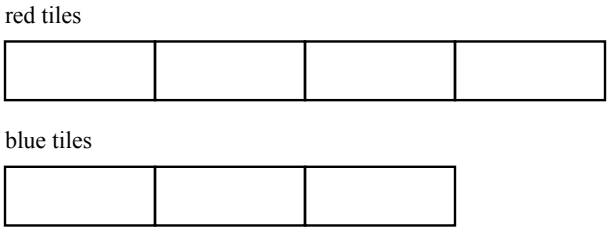
S. Return here near the end of the lesson

To keep a ratio the same, I ___ both quantities by ___.

Equivalent ratios

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. The diagram has 4 red tiles and 3 blue tiles. Write red : blue. Write blue : red.



P2. A trail mix uses 4 raisins for every 3 nuts. How many nuts go with 12 raisins?

Answer: _____

P3. A box has 20 red tiles and 15 blue tiles. Divide both counts by 5 to find an equivalent ratio.

Answer: _____

Equivalent ratios

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. A recipe uses 6 cups of oats for every 4 cups of seeds. How many cups of seeds go with 9 cups of oats?

Answer: _____

P5. Paint uses 2 cups yellow for every 5 cups blue. How much yellow goes with 15 cups blue?

Answer: _____

P6. Each game team has 3 builders for every 2 testers. How many builders go with 8 testers?

Answer: _____

Equivalent ratios

Follow the slide labels. P = problem; F = your turn; D = discussion.

F. A mix uses 5 cups oats for every 2 cups seeds. Ten cups oats makes __ batches and needs __ cups seeds.

Answer: _____

D. Mia changes 4 red : 3 blue to 7 red : 6 blue by adding 3 to each count. Is the ratio unchanged?

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. A bracelet pattern uses 7 green beads for every 4 white beads. How many green beads go with 12 white beads?

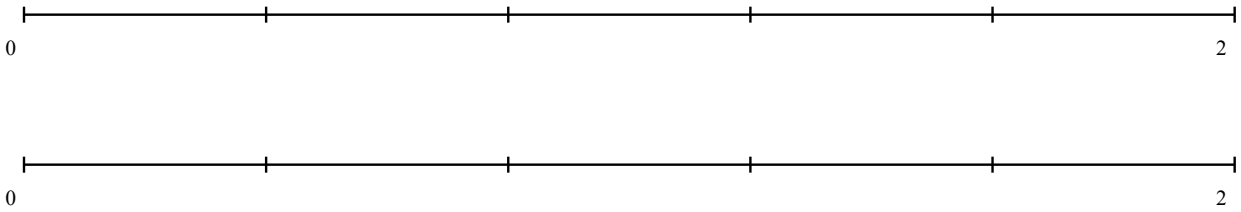
Answer: _____

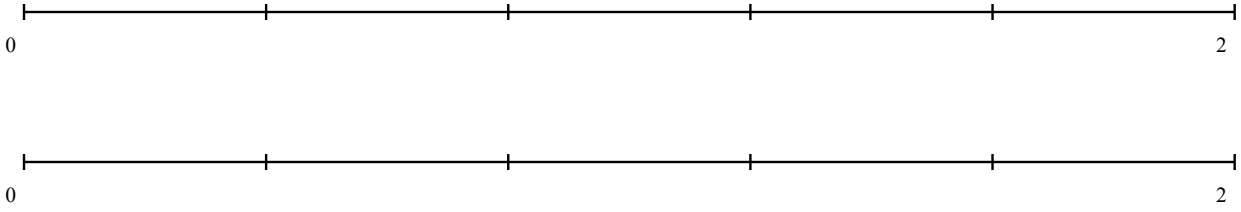
Ratio tables

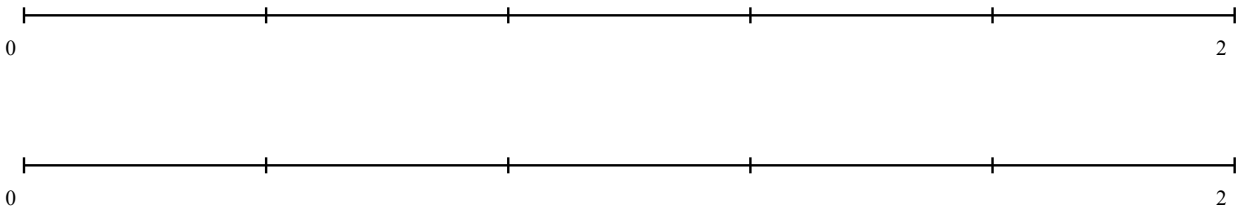
Quantity A: _____	Quantity B: _____

Quantity A: _____	Quantity B: _____

Double Number Line Template







Recall

Work from memory, then check with your tutor.

R1. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

Answer: _____

R2. Find $\frac{3}{4} - \frac{1}{3}$.

Answer: _____

R3. Is $\frac{2}{3} \times 9$ smaller or larger than 9? Compute to check.

Answer: _____

R4. Find $3 \times \frac{2}{3}$.

Answer: _____

S. Return here near the end of the lesson

To compare two mixtures fairly, I can make ___ the same.

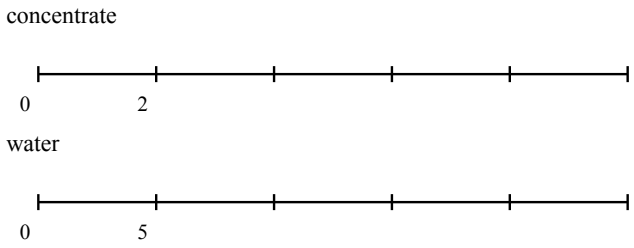
Ratio tables and comparisons

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Concentrate : water is 2 : 5. Complete pairs (2, 5), (4, __), and (6, 15).

concentrate	water
2	5
4	?
6	15

P2. For that 2 : 5 mixture, how much water matches 10 cups concentrate? Solve using a table and a double number line.



P3. Mix A uses 3 cups syrup to 4 cups water. Mix B uses 5 cups syrup to 8 cups water. Which has more syrup for the same water?

Answer: _____

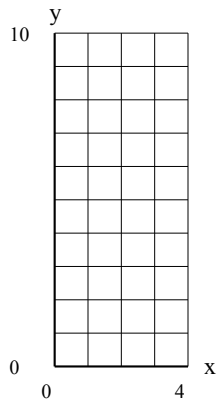
Ratio tables and comparisons

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Yellow : blue paint is 2 : 3. There are 35 cups in all. How much of each color?

Answer: _____

P5. Plot (1, 3), (2, 6), and (3, 9). Let x be bags and y be marbles. Does their straight line also pass through (0, 0)?



x: bags; y: marbles

F. Green : white is 4 : 1. There are 30 beads. Five equal parts means __ beads per part. Green = __; white = __.

Answer: _____

Ratio tables and comparisons

Follow the slide labels. P = problem; F = your turn; D = discussion.

D. Leo says 2 yellow : 3 blue means $\frac{2}{3}$ of all the paint is yellow. Repair his statement.

Answer: _____

Mixed review

M1. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

Answer: _____

M2. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

Answer: _____

M3. Which is closer to 1: $\frac{7}{8}$ or $\frac{4}{5}$?

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Mix A has 4 cups juice and 3 cups water. Mix B has 6 cups juice and 5 cups water. Which has the larger fraction of juice?

Answer: _____

Ratios and matching amounts

The big idea

A ratio compares two amounts in a chosen order. Equivalent ratios scale both amounts together. Tables and double number lines keep matching amounts connected.

Words to know

Ratio: A comparison of two amounts in a stated order. Example: 4 red to 3 blue is 4 : 3.

Equivalent ratios: Ratios that describe the same relationship. Example: 4 : 3 and 8 : 6 are equivalent.

Scale factor: The number used to multiply both amounts. Example: From 4 : 3 to 12 : 9, the factor is 3.

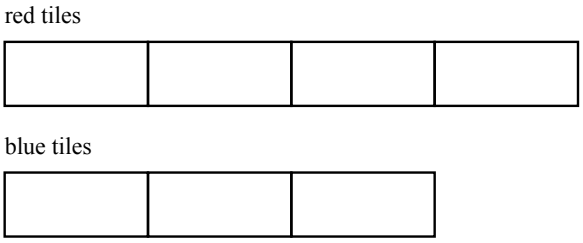
Ratio table: A table of matching amounts with the same ratio. Example: (2, 5), (4, 10), and (6, 15).

Double number line: Two aligned scales showing matching amounts. Example: Two cups concentrate matches five cups water.

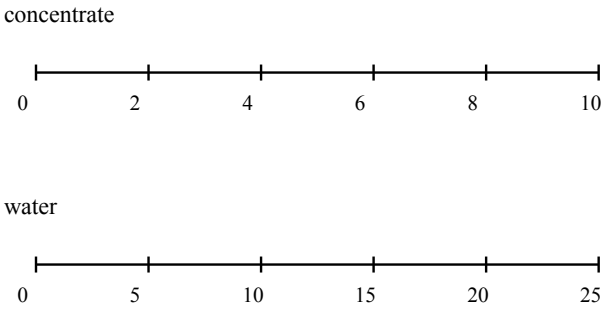
Part-to-whole: A comparison between one part and the total. Example: 2 red and 3 blue makes red $\frac{2}{5}$ of all tiles.

How it works

Repeat the whole 4 : 3 group. Equivalent ratios multiply or divide both quantities by the same factor.



Two cups concentrate pairs with five cups water. A ratio table records equal groups. Comparing mixtures needs a common amount or a common total.



Worked examples

1. A mix uses 3 cups oats for every 2 cups seeds. Find seeds for 12 cups oats.

12 ÷ 3 = 4 batches.

Each batch has 2 cups seeds.

4 × 2 = 8 cups seeds.

Answer: 8 cups seeds

2. Orange : purple tiles is 3 : 2. There are 25 tiles. Find each count.

3 + 2 = 5 equal parts.

25 ÷ 5 = 5 tiles per part.

Orange: 3 × 5 = 15. Purple: 2 × 5 = 10.

Answer: 15 orange, 10 purple

Common mistakes

Adds the same count to both colors. Why: Confuses additive and multiplicative change. Fix: Build identical batches.

Says 4 : 3 means four out of three. Why: Confuses part-to-part and part-to-whole. Fix: Count the total seven tiles before naming the red fraction.

Compares mixtures by subtraction. Why: A difference ignores mixture size. Fix: Use the same amount of water in both mixtures.

Try it

1. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

2. Is 6 : 10 equivalent to 9 : 15?

3. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

4. A double number line pairs 3 minutes with 8 laps. What pairs with 12 minutes?

5. Plot (1, 2), (2, 4), and (4, 8). Do they lie on a line through the origin?

Answers

1. 5 : 2; 2 : 5 2. Yes 3. 21 4. 32 laps 5. Yes

Where this connects

This builds on equal groups and fraction comparison. Next, unit rates and percents describe one-unit and hundred-unit comparisons.

Daily practice

A. Warm-up (3 minutes)

1. 4×3

Answer: _____

2. 4×7

Answer: _____

3. 4×4

Answer: _____

4. 4×9

Answer: _____

B. Practice

5. Compare $\frac{1}{3}$ and $\frac{1}{4}$.

Answer: _____

6. $2.5 + 0.25$

Answer: _____

7. 5×5

Answer: _____

8. A recipe uses 2 cups beans for 3 cups rice. How many cups beans go with $\frac{9}{2}$ cups rice?

Answer: _____

Practice and thinking

Built-in practice: Ratios and rates

9. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

Answer: _____

10. Is 6 : 10 equivalent to 9 : 15?

Answer: _____

11. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

Answer: _____

12. A team has 4 runners for every 3 walkers. There are 18 walkers. How many runners?

Answer: _____

C. Think

13. Mia changes 4 red : 3 blue to 7 red : 6 blue by adding 3 to each count. Is the ratio unchanged?

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Write $\frac{1}{2}$ as a decimal and a percent.

Answer: _____

2. Write $\frac{1}{4}$ as a decimal and a percent.

Answer: _____

3. Write $\frac{3}{4}$ as a decimal and a percent.

Answer: _____

4. Write $\frac{1}{8}$ as a decimal and a percent.

Answer: _____

B. Practice

5. Compare $\frac{1}{3}$ and $\frac{1}{4}$.

Answer: _____

6. 300 ÷ 100

Answer: _____

7. 72 ÷ 6

Answer: _____

8. Yellow : purple is 2 : 5. There are 28 beads. Find both counts.

Answer: _____

Practice and thinking

Built-in practice: Ratios and rates

9. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

Answer: _____

10. A double number line pairs 3 minutes with 8 laps. What pairs with 12 minutes?

Answer: _____

11. Plot (1, 2), (2, 4), and (4, 8). Do they lie on a line through the origin?

Answer: _____

12. A mix has 5 cups oats and 3 cups seeds. What fraction of the whole is seeds?

Answer: _____

C. Think

13. Leo says 2 yellow : 3 blue means $\frac{2}{3}$ of all the paint is yellow. Repair his statement.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. 4.2×10

Answer: _____

2. $400 \div 100$

Answer: _____

3. Double 4.5

Answer: _____

4. Half of 24

Answer: _____

B. Practice

5. Is 6 : 10 equivalent to 9 : 15?

Answer: _____

6. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

Answer: _____

7. Find $3 \times \frac{2}{3}$.

Answer: _____

8. Find $\frac{1}{2} + \frac{1}{3}$.

Answer: _____

Practice and thinking

B. Practice, continued

9. 4.5 + 0.25

Answer: _____

10. A box is 4 cm by 3 cm by 3 cm. Find its volume.

Answer: _____

11. 7 × 2

Answer: _____

12. A box is 7 cm by 2 cm by 3 cm. Find its volume.

Answer: _____

C. Think: weekly puzzle

13. A tile bag has red : blue = 3 : 2. The total is greater than 20 and less than 30. How many of each color?

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Round 448 to the nearest hundred.

Answer: _____

2. Round 4.72 to the nearest whole number.

Answer: _____

3. Use 20 × 5 to estimate 19.8 × 5.1.

Answer: _____

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

Answer: _____

B. Practice

5. Is 6 : 10 equivalent to 9 : 15?

Answer: _____

6. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

Answer: _____

7. Simplify the ratio 16 : 28 by dividing both counts by 4.

Answer: _____

8. Write a story for $\frac{1}{6} \div 3$ and solve it.

Answer: _____

Practice and thinking

Built-in practice: Fractions: compare and combine

9. Find $\frac{1}{2} + \frac{1}{3}$.

Answer: _____

10. Find $\frac{2}{3} + \frac{1}{4}$.

Answer: _____

11. $\frac{4}{3} - \frac{3}{4}$

Answer: _____

12. $\frac{3}{2} - \frac{3}{4}$

Answer: _____

C. Think

13. A team has 4 runners for every 3 walkers. There are 18 walkers. How many runners?

Answer: _____

14. Leo says 2 yellow : 3 blue means $\frac{2}{3}$ of all the paint is yellow. Repair his statement.

Answer: _____

Weekly check

20 minutes. No calculator. Show your work.

1. Is 6 : 10 equivalent to 9 : 15?

Answer: _____

2. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

Answer: _____

3. Simplify the ratio 16 : 28 by dividing both counts by 4.

Answer: _____

4. Simplify the ratio 8 : 14 by dividing both counts by 2.

Answer: _____

5. Compare $\frac{2}{4}$ and $\frac{2}{5}$.

Answer: _____

Weekly check

6. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

Answer: _____

7. Find $\frac{3}{4} \times \frac{4}{5}$.

Answer: _____

8. 6.5 + 0.25

Answer: _____

9. A box is 6 cm by 5 cm by 3 cm. Find its volume.

Answer: _____

10. Mia changes 4 red : 3 blue to 7 red : 6 blue by adding 3 to each count. Is the ratio unchanged?

Answer: _____

Recall

Work from memory, then check with your tutor.

R1. Find $\frac{2}{3} + \frac{1}{4}$.

Answer: _____

R2. Yellow : purple is 2 : 5. There are 28 beads. Find both counts.

Answer: _____

R3. Share $\frac{1}{3}$ liter equally among 2 cups. How much per cup?

Answer: _____

R4. A recipe uses 2 cups beans for 3 cups rice. How many cups beans go with $\frac{9}{2}$ cups rice?

Answer: _____

S. Return here near the end of the lesson

The word per tells me to find the amount for one ___.

Rates and unit rates

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. A rider goes 20 kilometers in 4 hours at a steady speed. Find kilometers per hour.

Answer: _____

P2. Eight apples cost $6.00. Find dollars per apple.

Answer: _____

P3. Two cups of rice make 5 servings. Find cups per serving.

Answer: _____

Rates and unit rates

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. A printer makes 15 cards in 3 minutes. Find cards per minute and minutes per card.

Answer: _____

P5. Pack A has 6 pencils for $2.40. Pack B has 10 pencils for $3.50. Which costs less per pencil?

Answer: _____

P6. A skater travels 3 miles in 20 minutes at steady speed. Find miles per hour.

Answer: _____

Rates and unit rates

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. A robot rolls 4 meters each second. Complete distances for 0, 2, and 5 seconds.

Answer: _____

F. A scooter travels 5 miles in 30 minutes. There are __ half-hours in one hour, so the speed is __ miles per hour.

Answer: _____

D. Ava says 8 apples for \$6 costs $\frac{4}{3}$ dollars per apple because $8 \div 6 = \frac{4}{3}$. Is her unit correct?

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. A cyclist travels 9 miles in 45 minutes at steady speed. Find miles per hour and minutes per mile.

Answer: _____

Recall

Work from memory, then check with your tutor.

R1. Write a story for $\frac{1}{6} \div 3$ and solve it.

Answer: _____

R2. A team has 4 runners for every 3 walkers. There are 18 walkers. How many runners?

Answer: _____

R3. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

Answer: _____

R4. A double number line pairs 3 minutes with 8 laps. What pairs with 12 minutes?

Answer: _____

S. Return here near the end of the lesson

Before using a percent, I decide which amount represents ___.

Percent as a rate per hundred

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Find 10%, 25%, 50%, and 75% of 80 counters.

Answer: _____

P2. A shelf holds 40 books. Twelve are mysteries. What percent are mysteries?

Answer: _____

P3. Twenty-four stickers are 40% of a collection. How many stickers are in the whole collection?

Answer: _____

Percent as a rate per hundred

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Find 125% of 48 minutes.

Answer: _____

P5. Find 35% of 60 tiles.

Answer: _____

P6. Find 15% of $18.00.

Answer: _____

Percent as a rate per hundred

Follow the slide labels. P = problem; F = your turn; D = discussion.

F. Twelve coins are 20% of a jar. Ten percent is __ coins. One hundred percent is __ coins.

Answer: _____

D. Omar says 24 is 40% of 9.6 because $0.40 \times 24 = 9.6$. Explain the error.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Twenty-one marbles are 35% of a bag. How many marbles are in the bag?

Answer: _____

Mini Check A

15 minutes. No calculator. Show your work.

1. Compare $\frac{2}{9}$ and $\frac{7}{9}$.

Answer: _____

2. Which is closer to 1: $\frac{7}{8}$ or $\frac{4}{5}$?

Answer: _____

3. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

Answer: _____

4. Is 6 : 10 equivalent to 9 : 15?

Answer: _____

5. A bus covers 10 miles in 25 minutes at constant speed. How many miles in one hour?

Answer: _____

Mini Check A

6. Pack A has 4 erasers for $1.80. Pack B has 6 for $2.40. Which costs less per eraser?

Answer: _____

7. Find 140% of 30 points.

Answer: _____

8. Find 35% of 80 tiles.

Answer: _____

Rates and percents

The big idea

A unit rate describes an amount per one unit. Percent means per hundred. Label the whole before deciding whether to find a part, a whole, or a percent.

Words to know

Rate: A ratio comparing quantities with different units. Example: 15 kilometers in 3 hours.

Unit rate: An amount per one unit of another quantity. Example: 5 kilometers per hour.

Per: For each one. Example: $0.60 per apple means $0.60 for one apple.

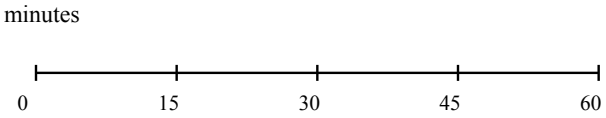
Percent: A rate per hundred. Example: 35% means 35 for every 100.

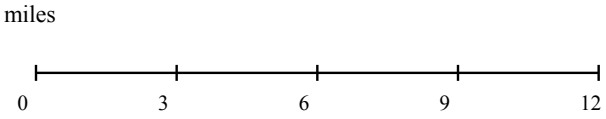
Whole: The amount that represents 100%. Example: If all 80 tiles count, 80 is the whole.

Benchmark percent: A familiar percent useful for reasoning. Example: 25% is one fourth.

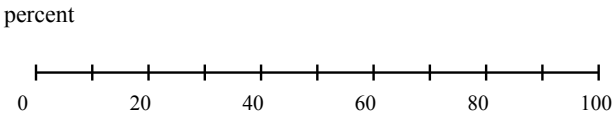
How it works

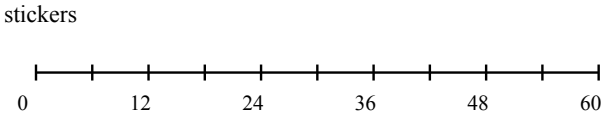
Four 15-minute intervals make one hour. A unit rate names how much of one quantity goes with one unit of another. Its units show the division order.





100% labels the unknown whole. Percent compares an amount with a whole split into 100 equal parts.





Worked examples

1. A walker travels 2 miles in 30 minutes. Find miles per hour.

60 minutes is two groups of 30.

Two groups of 2 miles make 4 miles.

The rate is 4 miles per hour.

Answer: 4 miles per hour

2. Eighteen blocks are 30% of a tower set. Find the whole set.

30% has three 10% groups.

18 ÷ 3 = 6 blocks per 10%.

Ten groups give 10 × 6 = 60 blocks.

Answer: 60 blocks

Common mistakes

Divides in the reverse direction. Why: Does not track the per unit. Fix: Read the desired rate aloud before dividing.

Treats 20 minutes as 0.20 hours. Why: Uses base ten instead of 60 minutes per hour. Fix: Partition an hour into three 20-minute sections.

Multiplies the known part by the percent. Why: Treats every percent problem as finding a part. Fix: Place the known part and its percent on aligned lines.

Try it

1. A bus covers 10 miles in 25 minutes at constant speed. How many miles in one hour?

2. Pack A has 4 erasers for $1.80. Pack B has 6 for $2.40. Which costs less per eraser?

3. Find 140% of 30 points.

4. Find 35% of 80 tiles.

5. Find 12% of $15.00.

Answers

1. 24 miles 2. Pack B: $0.40 each 3. 42 points 4. 28 tiles 5. 1.8000 dollars

Where this connects

This builds on equivalent ratios. Fraction division later extends the same grouping and scaling ideas.

Daily practice

A. Warm-up (3 minutes)

1. 5×3

Answer: _____

2. 5×7

Answer: _____

3. 5×4

Answer: _____

4. 5×9

Answer: _____

B. Practice

5. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

Answer: _____

6. $3.5 + 0.25$

Answer: _____

7. 6×6

Answer: _____

8. A walker travels 12 miles in 3 hours. Find miles per hour.

Answer: _____

Practice and thinking

Built-in practice: Ratios and rates

9. A bus covers 10 miles in 25 minutes at constant speed. How many miles in one hour?

Answer: _____

10. Pack A has 4 erasers for $1.80. Pack B has 6 for $2.40. Which costs less per eraser?

Answer: _____

11. A machine makes 24 labels in 6 minutes. Find labels per minute and minutes per label.

Answer: _____

12. A snail moves 3 centimeters each minute. Find the distance after 0, 4, and 7 minutes.

Answer: _____

C. Think

13. Ava says 8 apples for $6 costs $\frac{4}{3}$ dollars per apple because $8 \div 6 = \frac{4}{3}$. Is her unit correct?

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Write $\frac{1}{2}$ as a decimal and a percent.

Answer: _____

2. Write $\frac{1}{4}$ as a decimal and a percent.

Answer: _____

3. Write $\frac{3}{4}$ as a decimal and a percent.

Answer: _____

4. Write $\frac{1}{8}$ as a decimal and a percent.

Answer: _____

B. Practice

5. A mix has 5 cups oats and 3 cups seeds. What fraction of the whole is seeds?

Answer: _____

6. A box is 4 cm by 3 cm by 3 cm. Find its volume.

Answer: _____

7. 84 ÷ 7

Answer: _____

8. Find 10%, 50%, and 75% of 40 counters.

Answer: _____

Practice and thinking

Built-in practice: Percent practice

9. Find 140% of 30 points.

Answer: _____

10. Find 35% of 80 tiles.

Answer: _____

11. 12 shells are 30% of a collection. Find the whole collection.

Answer: _____

12. 9 of 12 tiles are green. What percent are green?

Answer: _____

C. Think

13. Omar says 24 is 40% of 9.6 because $0.40 \times 24 = 9.6$. Explain the error.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. 5.2 × 10

Answer: _____

2. 500 ÷ 100

Answer: _____

3. Double 5.5

Answer: _____

4. Half of 30

Answer: _____

B. Practice

5. A walker travels 12 miles in 3 hours. Find miles per hour.

Answer: _____

6. 9 of 12 tiles are green. What percent are green?

Answer: _____

7. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

Answer: _____

8. A line marks 0, 3, and 6. How far from 3 toward 6 is $4\frac{1}{2}$?

Answer: _____

Practice and thinking

B. Practice, continued

9. $\frac{1}{4} + \frac{1}{4}$

Answer: _____

10. 500 ÷ 100

Answer: _____

11. 8 × 3

Answer: _____

12. A box is 8 cm by 3 cm by 3 cm. Find its volume.

Answer: _____

C. Think: weekly puzzle

13. A mystery number has this property: 25% of it is 6 more than 10% of it. Find the number.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Round 548 to the nearest hundred.

Answer: _____

2. Round 5.72 to the nearest whole number.

Answer: _____

3. Use 20×5 to estimate 19.8×5.1.

Answer: _____

4. Use $60 \div 3$ to estimate $59.4 \div 3.1$.

Answer: _____

B. Practice

5. A walker travels 12 miles in 3 hours. Find miles per hour.

Answer: _____

6. 9 of 12 tiles are green. What percent are green?

Answer: _____

7. 12 shells are 30% of a collection. Find the whole collection.

Answer: _____

8. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

Answer: _____

Practice and thinking

Built-in practice: Decimal practice

9. 2.4 + 1.75

Answer: _____

10. 3.4 × 0.5

Answer: _____

11. 4.4 + 1.75

Answer: _____

12. 5.4 × 0.5

Answer: _____

C. Think

13. A walker travels 16 miles in 4 hours. Find miles per hour.

Answer: _____

14. Omar says 24 is 40% of 9.6 because 0.40 × 24 = 9.6. Explain the error.

Answer: _____

Weekly check

20 minutes. No calculator. Show your work.

1. A walker travels 12 miles in 3 hours. Find miles per hour.

Answer: _____

2. 9 of 12 tiles are green. What percent are green?

Answer: _____

3. 12 shells are 30% of a collection. Find the whole collection.

Answer: _____

4. A walker travels 8 miles in 2 hours. Find miles per hour.

Answer: _____

5. Is $\frac{2}{3} \times 9$ smaller or larger than 9? Compute to check.

Answer: _____

Weekly check

6. How many $\frac{1}{4}$-meter pieces fit in 3 meters?

Answer: _____

7. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

Answer: _____

8. $\frac{1}{6} + \frac{1}{6}$

Answer: _____

9. $700 \div 100$

Answer: _____

10. Ava says 8 apples for \$6 costs $\frac{4}{3}$ dollars per apple because $8 \div 6 = \frac{4}{3}$. Is her unit correct?

Answer: _____

Recall

Work from memory, then check with your tutor.

R1. Which is closer to 1: $\frac{7}{8}$ or $\frac{4}{5}$?

Answer: _____

R2. Red : blue is 3 : 5. How many blue tiles go with 12 red tiles?

Answer: _____

R3. 5 markers cost $3.25. Find dollars per marker.

Answer: _____

R4. Find both $4 \div \frac{1}{5}$ and $\frac{1}{5} \div 4$.

Answer: _____

S. Return here near the end of the lesson

To divide by a fraction, I ask how many ___-sized groups fit in ___.

The meaning of fraction division

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Four meters are cut into $\frac{1}{5}$-meter pieces. How many pieces?

Answer: _____

P2. How many $\frac{2}{3}$-cup scoops fit in 4 cups?

Answer: _____

P3. How many $\frac{1}{8}$-meter pieces fit in $\frac{3}{4}$ meter?

Answer: _____

The meaning of fraction division

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. How many $\frac{2}{3}$-cup groups fit in $\frac{5}{6}$ cup? A fraction of a group is allowed.

Answer: _____

P5. What fraction of a $\frac{5}{6}$-cup scoop is filled by $\frac{1}{3}$ cup?

Answer: _____

P6. A jar is $\frac{2}{3}$ full after $\frac{3}{5}$ liter is poured in. How many liters fill the whole jar?

Answer: _____

The meaning of fraction division

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. Write a story for $\frac{3}{4} \div \frac{1}{2}$. Solve it.

Answer: _____

F. Find $\frac{5}{8} \div \frac{1}{4}$. One group is __ eighths. Five eighths holds __ groups.

Answer: _____

D. Ben says $\frac{1}{3} \div \frac{5}{6} = \frac{5}{18}$. He multiplies the fractions. Explain what division asks here.

Answer: _____

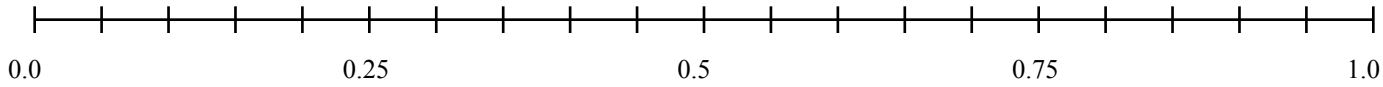
Exit ticket

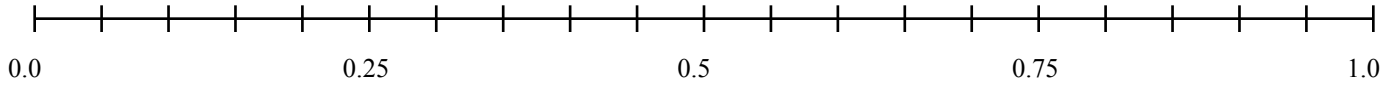
Work on your own. Show a model or reason when the question asks for one.

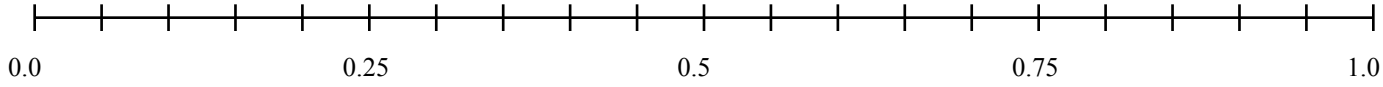
E. How much of a $\frac{3}{4}$-meter length is $\frac{2}{5}$ meter? Draw equal parts to show $\frac{2}{5} \div \frac{3}{4}$.

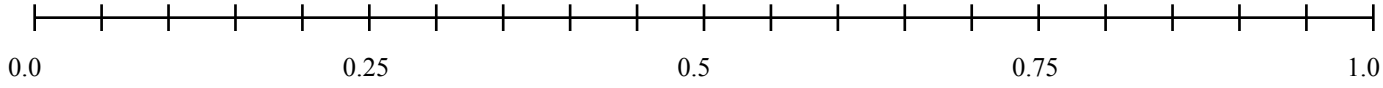
Answer: _____

0 to 1

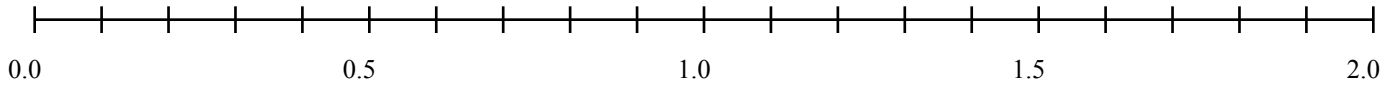


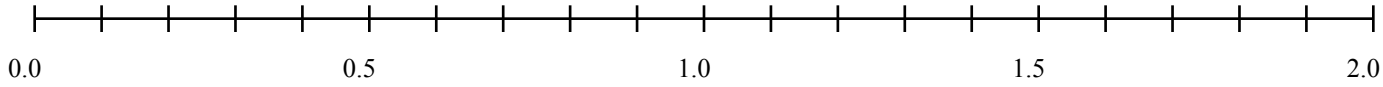


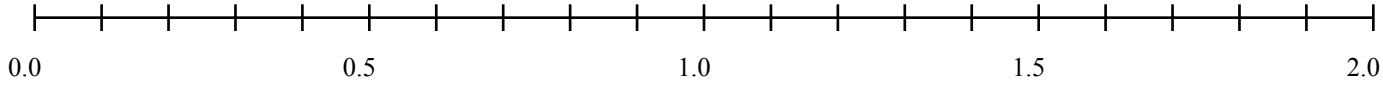


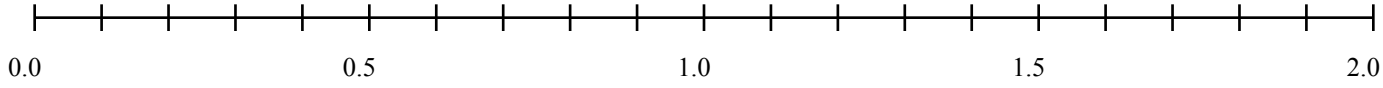


0 to 2









Recall

Work from memory, then check with your tutor.

R1. Find $\frac{1}{2} + \frac{1}{3}$.

Answer: _____

R2. Complete the ratio-table pair: (3, 4), (9, __).

Answer: _____

R3. 9 of 12 tiles are green. What percent are green?

Answer: _____

R4. How many $\frac{1}{4}$-meter pieces fit in 3 meters?

Answer: _____

S. Return here near the end of the lesson

I can check a fraction quotient by multiplying it by ___.

Fraction division and reciprocals

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Share $\frac{3}{4}$ liter equally among 5 bottles. How much per bottle?

Answer: _____

P2. How many $\frac{3}{5}$-cup groups fit in 4 cups? Include a fraction of a group.

Answer: _____

P3. Find $\frac{7}{8} \div \frac{1}{4}$. Check by multiplying the quotient by $\frac{1}{4}$.

Answer: _____

Fraction division and reciprocals

Follow the slide labels. P = problem; F = your turn; D = discussion.

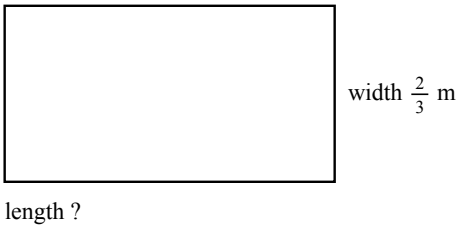
P4. Find $1\ \frac{7}{8} \div \frac{5}{8}$.

Answer: _____

P5. A cord is $3\ \frac{1}{2}$ meters long. Each piece is $\frac{7}{8}$ meter. How many pieces?

Answer: _____

P6. A rectangle has area $\frac{5}{6}$ square meter and width $\frac{2}{3}$ meter. Find its length.



Fraction division and reciprocals

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. A box has edges $\frac{2}{3}$, $\frac{3}{4}$, and 2 meters. Find its volume.

Answer: _____

F. Complete: $\frac{4}{5} \div \frac{2}{3} = \frac{4}{5} \times __ = __$. Check by multiplying by $\frac{2}{3}$.

Answer: _____

D. Zoe writes $\frac{7}{8} \div \frac{1}{4} = \frac{8}{7} \times \frac{1}{4} = \frac{2}{7}$. Explain which number should be inverted.

Answer: _____

Mixed review

M1. Find $3 \times \frac{2}{3}$.

Answer: _____

M2. A recipe uses 2 cups beans for 3 cups rice. How many cups beans go with $\frac{9}{2}$ cups rice?

Answer: _____

M3. 3 markers cost $1.95. Find dollars per marker.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. A ribbon is $2\frac{5}{8}$ yards long. Each piece is $\frac{3}{8}$ yard. How many pieces can be cut?

Answer: _____

Dividing fractions

The big idea

Division asks how many divisor-sized groups fit. Equal small units make the comparison visible. The reciprocal rule records that reasoning efficiently.

Words to know

Dividend: The amount being divided. Example: In $2 \div \frac{3}{4}$, the dividend is 2.

Divisor: The amount that names one group in a grouping problem. Example: In $2 \div \frac{3}{4}$, one group is $\frac{3}{4}$.

Quotient: The result of division. Example: $\frac{3}{4} \div \frac{1}{4} = 3$ groups.

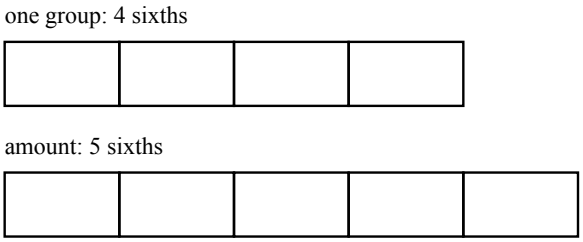
Reciprocal: A number that multiplies with another number to make 1. Example: $\frac{3}{5}$ and $\frac{5}{3}$ are reciprocals.

Mixed number: A whole number plus a fraction. Example: $2\ \frac{1}{3} = \frac{7}{3}$.

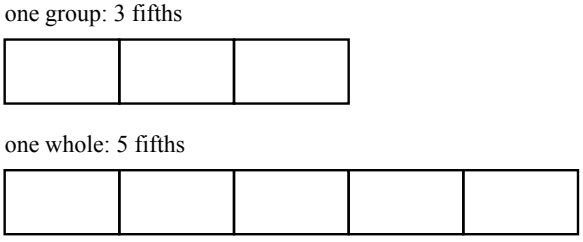
Volume: The amount of three-dimensional space a solid fills. Example: A box uses cubic units.

How it works

Five sixths is five fourths of a four-sixths group. The divisor names one group. A quotient tells how many of those groups fit, even when it is part of a group.



One whole has $\frac{5}{3}$ groups of size $\frac{3}{5}$. Dividing by a fraction counts groups. Multiplying by its reciprocal gives the same count.



Worked examples

1. Find $\frac{3}{4} \div \frac{1}{2}$ with a tape.

Rename one half as two fourths.

Three fourths contains one full pair of fourths and one fourth left.

The remaining fourth is half of a two-fourths group.

Answer: $1\ \frac{1}{2}$

2. Find $\frac{2}{3} \div \frac{3}{5}$.

One whole contains $\frac{5}{3}$ groups of size $\frac{3}{5}$.

Two thirds of a whole contains $\frac{2}{3} \times \frac{5}{3}$ groups.

Multiply to get $\frac{10}{9}$, or $1\ \frac{1}{9}$.

Answer: $1\ \frac{1}{9}$

Common mistakes

Reverses dividend and divisor. Why: Has not named the available amount. Fix: Label amount first and group size second.

Multiplies the fractions. Why: Applies a rule before making sense of division. Fix: Count equal pieces before writing a rule.

Inverts the dividend. Why: Memorizes a rule without a division meaning. Fix: Check quotient × divisor = dividend.

Try it

1. How many $\frac{3}{4}$-liter groups fit in $\frac{7}{8}$ liter? Include part of a group.

2. A bottle is $\frac{3}{4}$ full and holds $\frac{2}{3}$ liter so far. Find its full capacity.

3. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

4. A ribbon is $4\ \frac{1}{2}$ meters long. Pieces measure $\frac{3}{8}$ meter. How many pieces fit?

5. Find $\frac{4}{7} \div \frac{2}{5}$. Check the result by multiplying by $\frac{2}{5}$.

Answers

1. $1\ \frac{1}{6}$ groups 2. $\frac{8}{9}$ liter 3. $\frac{5}{24}$ liter 4. 12 pieces 5. $1\ \frac{3}{7}$

Where this connects

This builds on fraction multiplication and equivalent ratios. Decimal division will also preserve group counts by renaming units.

Daily practice

A. Warm-up (3 minutes)

1. 6×3

Answer: _____

2. 6×7

Answer: _____

3. 6×4

Answer: _____

4. 6×9

Answer: _____

B. Practice

5. Find 10%, 50%, and 75% of 40 counters.

Answer: _____

6. Find both $4 \div \frac{1}{5}$ and $\frac{1}{5} \div 4$.

Answer: _____

7. 7×2

Answer: _____

8. Find $\frac{1}{2} \div \frac{3}{4}$.

Answer: _____

Practice and thinking

Built-in practice: Fraction operations

9. Find $\frac{1}{2} \div \frac{4}{5}$.

Answer: _____

10. $\frac{2}{3} \div \frac{1}{5}$

Answer: _____

11. $1 \div \frac{3}{5}$

Answer: _____

12. $\frac{5}{3} \div \frac{1}{5}$

Answer: _____

C. Think

13. Ben says $\frac{1}{3} \div \frac{5}{6} = \frac{5}{18}$. He multiplies the fractions. Explain what division asks here.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Write $\frac{1}{2}$ as a decimal and a percent.

Answer: _____

2. Write $\frac{1}{4}$ as a decimal and a percent.

Answer: _____

3. Write $\frac{3}{4}$ as a decimal and a percent.

Answer: _____

4. Write $\frac{1}{8}$ as a decimal and a percent.

Answer: _____

B. Practice

5. Find 140% of 30 points.

Answer: _____

6. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

Answer: _____

7. $96 \div 8$

Answer: _____

8. Find $3\ \frac{1}{2} \div \frac{3}{4}$.

Answer: _____

Practice and thinking

Built-in practice: Fraction operations

9. Which is the reciprocal of $2\frac{1}{2}$: $\frac{2}{5}$ or $\frac{5}{2}$? Explain using a product of 1.

Answer: _____

10. Find $1\frac{1}{2} \div \frac{3}{4}$.

Answer: _____

11. $\frac{4}{3} \div \frac{3}{5}$

Answer: _____

12. $\frac{4}{3} \times \frac{4}{5}$

Answer: _____

C. Think

13. Zoe writes $\frac{7}{8} \div \frac{1}{4} = \frac{8}{7} \times \frac{1}{4} = \frac{2}{7}$. Explain which number should be inverted.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. 6.2×10

Answer: _____

2. $600 \div 100$

Answer: _____

3. Double 6.5

Answer: _____

4. Half of 36

Answer: _____

B. Practice

5. Find $3 \div \frac{2}{3}$.

Answer: _____

6. Find $\frac{1}{2} \div \frac{4}{5}$.

Answer: _____

7. Complete the ratio-table pair: (3, 4), (12, __).

Answer: _____

8. A recipe uses 2 cups beans for 3 cups rice. How many cups beans go with $\frac{15}{2}$ cups rice?

Answer: _____

Practice and thinking

B. Practice, continued

9. Find $\frac{1}{2} + \frac{1}{3}$.

Answer: _____

10. Find $\frac{3}{4} \times \frac{4}{5}$.

Answer: _____

11. 2×4

Answer: _____

12. A box is 2 cm by 4 cm by 3 cm. Find its volume.

Answer: _____

C. Think: weekly puzzle

13. A jug holds 2 liters. You have $\frac{1}{3}$-liter and $\frac{1}{2}$-liter scoops. Use both sizes and exactly five full scoops to fill it. How many of each?

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Round 648 to the nearest hundred.

Answer: _____

2. Round 6.72 to the nearest whole number.

Answer: _____

3. Use 20×5 to estimate 19.8×5.1.

Answer: _____

4. Use $60 \div 3$ to estimate $59.4 \div 3.1$.

Answer: _____

B. Practice

5. Find $3 \div \frac{2}{3}$.

Answer: _____

6. Find $\frac{1}{2} \div \frac{4}{5}$.

Answer: _____

7. Jules says $\frac{1}{2} \div \frac{2}{3} = \frac{1}{3}$. Find and fix the error.

Answer: _____

8. Plot (1, 2), (2, 4), and (4, 8). Do they lie on a line through the origin?

Answer: _____

Practice and thinking

Built-in practice: Percent practice

9. Find 12% of $15.00.

Answer: _____

10. 18 shells are 30% of a collection. Find the whole collection.

Answer: _____

11. 12 of 16 tiles are green. What percent are green?

Answer: _____

12. Find 25% of 40 stickers.

Answer: _____

C. Think

13. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

Answer: _____

14. Zoe writes $\frac{7}{8} \div \frac{1}{4} = \frac{8}{7} \times \frac{1}{4} = \frac{2}{7}$. Explain which number should be inverted.

Answer: _____

Weekly check

20 minutes. No calculator. Show your work.

1. Find $3 \div \frac{2}{3}$.

Answer: _____

2. Find $\frac{1}{2} \div \frac{4}{5}$.

Answer: _____

3. Jules says $\frac{1}{2} \div \frac{2}{3} = \frac{1}{3}$. Find and fix the error.

Answer: _____

4. A rectangle has area $\frac{5}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

Answer: _____

5. 3 markers cost $1.95. Find dollars per marker.

Answer: _____

Weekly check

6. Find 10%, 50%, and 75% of 40 counters.

Answer: _____

7. Complete the ratio-table pair: (3, 4), (15, __).

Answer: _____

8. Find $\frac{1}{2} + \frac{1}{3}$.

Answer: _____

9. Find $\frac{3}{4} \times \frac{4}{5}$.

Answer: _____

10. Ben says $\frac{1}{3} \div \frac{5}{6} = \frac{5}{18}$. He multiplies the fractions. Explain what division asks here.

Answer: _____

Recall

Work from memory, then check with your tutor.

R1. Complete the ratio-table pair: (3, 4), (12, __).

Answer: _____

R2. 18 shells are 30% of a collection. Find the whole collection.

Answer: _____

R3. A bottle is $\frac{3}{4}$ full and holds $\frac{2}{3}$ liter so far. Find its full capacity.

Answer: _____

R4. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

Answer: _____

S. Return here near the end of the lesson

I check a decimal answer by naming the units and ___.

Decimal place value and products

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Find 4.7 + 8.036.

Answer: _____

P2. Find 12 − 4.685.

Answer: _____

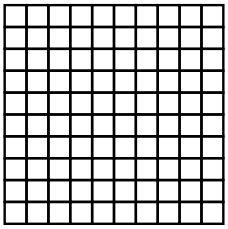
P3. A rope costs $2.35 per meter. What do 4 meters cost?

Answer: _____

Decimal place value and products

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Find 0.6 × 0.4. Is the product less than each factor?



P5. Find 0.15 × 0.02.

Answer: _____

P6. A notebook costs $3.85 and a pen costs $1.67. You pay $10. Find the change.

Answer: _____

Decimal place value and products

Follow the slide labels. P = problem; F = your turn; D = discussion.

F. Complete: 0.3 × 0.07 means 3 tenths × 7 hundredths = 21 __ = __.

Answer: _____

D. Isaac says 0.6 × 0.4 = 2.4 because 6 × 4 = 24. Use place value to repair this.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Find 0.45 × 0.06. Explain why your decimal position is reasonable.

Answer: _____

Recall

Work from memory, then check with your tutor.

R1. Red : blue is 3 : 5. How many blue tiles go with 6 red tiles?

Answer: _____

R2. A snail moves 3 centimeters each minute. Find the distance after 0, 4, and 7 minutes.

Answer: _____

R3. Find $\frac{3}{5} \div \frac{1}{5}$.

Answer: _____

R4. Complete the ratio-table pair: (3, 4), (12, __).

Answer: _____

S. Return here near the end of the lesson

To keep a quotient unchanged, I multiply both ___ and ___ by the same number.

Long division with decimals

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Use long division to find 672 ÷ 24.

Answer: _____

P2. Find 27 ÷ 8. Continue into decimal places.

Answer: _____

P3. Find 8.4 ÷ 7.

Answer: _____

Long division with decimals

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Find 6.3 ÷ 0.09 by writing an equivalent division with a whole-number divisor.

Answer: _____

P5. Find 7 ÷ 20.

Answer: _____

P6. A 3.6-meter ribbon is cut into 0.4-meter pieces. How many pieces?

Answer: _____

Long division with decimals

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. Find 1,176 ÷ 28 and check by multiplication.

Answer: _____

F. Complete: 5.4 ÷ 0.03 = __ ÷ 3 = __.

Answer: _____

D. Lina changes 6.3 ÷ 0.09 into 6.3 ÷ 9 and gets 0.7. Which change is missing?

Answer: _____

Mixed review

M1. A mix has 5 cups oats and 3 cups seeds. What fraction of the whole is seeds?

Answer: _____

M2. Find 12% of $15.00.

Answer: _____

M3. Find $2\frac{1}{2} \div \frac{3}{4}$.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Find 8.4 ÷ 0.07. Write an equivalent division and check by multiplication.

Answer: _____

Decimals and long division

The big idea

Decimal digits name parts of a whole. Addition and subtraction combine equal place-value units. Division preserves its quotient when both quantities scale equally.

Words to know

Place value: A digit’s value depends on its position. Example: The 4 in 0.04 means four hundredths.

Estimate: A nearby value that helps check an answer. Example: 6.2×3 is close to $6 \times 3 = 18$.

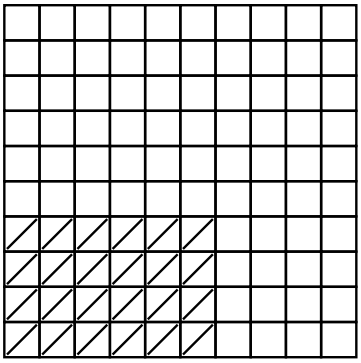
Thousandth: One of 1,000 equal parts of a whole. Example: 0.007 means seven thousandths.

Remainder: The amount left after equal groups are made. Example: $17 \div 5$ gives 3 groups with 2 left.

Equivalent division: A new dividend and divisor with the same quotient. Example: $6.4 \div 0.8 = 64 \div 8$.

Decimal quotient: A division result with fractional places written in decimals. Example: $7 \div 2 = 3.5$.

How it works



$0.6 \times 0.4 = 24$ hundredths. Addition combines like-sized units. Multiplication combines both the number of parts and the size of each part.

dividend	divisor
4.8	0.06
480	6

Multiply both quantities by 100. The quotient stays 80. Multiplying the dividend and divisor by the same power of ten preserves the quotient.

Worked examples

1. Find 3.8 + 0.475.

Write 3.800 + 0.475.

Add thousandths to thousandths and tenths to tenths.

3.800 + 0.475 = 4.275. The sum is a little above 4.

Answer: 4.275

2. Find 4.8 ÷ 0.06.

Rename both quantities in hundredths.

4.8 is 480 hundredths; 0.06 is 6 hundredths.

480 ÷ 6 = 80. Check 80 × 0.06 = 4.8.

Answer: 80

3. Long division: 936 ÷ 24

24 fits into 93 three times. Write 3 above the tens place. Subtract 72 from 93 to leave 21. Bring down the 6 to make 216. 24 fits into 216 nine times. Write 9 above the ones place. Subtract 216 to leave 0. The quotient is 39. Check: 39 × 24 = 936.

Common mistakes

Aligns the rightmost digits. Why: Treats every digit as the same unit. Fix: Write zeros to make place-value columns visible.

Makes a product larger than both small factors. Why: Keeps whole-number multiplication intuition. Fix: Take a part of one shaded part of a unit square.

Moves only one decimal point. Why: Uses a memorized instruction without equivalence. Fix: Rename both quantities in the same unit.

Try it

1. Three notebooks cost $2.45 each. What is the total cost?

2. A ball costs $4.65 and a card costs $2.78. Find the change from $10.

3. Find $9.6 \div 8$.

4. Find $9 \div 40$.

5. Find $1{,}344 \div 32$. Verify by multiplication.

Answers

1. $7.35 2. $2.57 3. 1.2 4. 0.225 5. 42

Where this connects

This builds on whole-number operations and fraction groups. Accurate decimal arithmetic supports expressions and equations next week.

Daily practice

A. Warm-up (3 minutes)

1. 7×3

Answer: _____

2. 7×7

Answer: _____

3. 7×4

Answer: _____

4. 7×9

Answer: _____

B. Practice

5. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

Answer: _____

6. A double number line pairs 3 minutes with 8 laps. What pairs with 12 minutes?

Answer: _____

7. 8×3

Answer: _____

8. Find $1.45 + 3.078$.

Answer: _____

Practice and thinking

Built-in practice: Decimal practice

9. Find 2.45 + 4.078.

Answer: _____

10. Find 0.25 × 0.02.

Answer: _____

11. Find 10 − 2.286.

Answer: _____

12. Find 0.8 × 0.7. Is the product below both factors?

Answer: _____

C. Think

13. Isaac says 0.6 × 0.4 = 2.4 because 6 × 4 = 24. Use place value to repair this.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Write $\frac{1}{2}$ as a decimal and a percent.

Answer: _____

2. Write $\frac{1}{4}$ as a decimal and a percent.

Answer: _____

3. Write $\frac{3}{4}$ as a decimal and a percent.

Answer: _____

4. Write $\frac{1}{8}$ as a decimal and a percent.

Answer: _____

B. Practice

5. 18 shells are 30% of a collection. Find the whole collection.

Answer: _____

6. Find $\frac{3}{4} - \frac{1}{3}$.

Answer: _____

7. $24 \div 2$

Answer: _____

8. Find $1.4 \div 0.07$.

Answer: _____

Practice and thinking

Built-in practice: Multiplication and long division

9. Find 312 ÷ 24.

Answer: _____

10. 117 × 12

Answer: _____

11. Find 1,344 ÷ 32. Verify by multiplication.

Answer: _____

12. 52 ÷ 4

Answer: _____

C. Think

13. Lina changes 6.3 ÷ 0.09 into 6.3 ÷ 9 and gets 0.7. Which change is missing?

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. 7.2 × 10

Answer: _____

2. 700 ÷ 100

Answer: _____

3. Double 7.5

Answer: _____

4. Half of 42

Answer: _____

B. Practice

5. Find 2.1 ÷ 0.07.

Answer: _____

6. Find 0.25 × 0.03.

Answer: _____

7. Find 12% of $15.00.

Answer: _____

8. 24 shells are 30% of a collection. Find the whole collection.

Answer: _____

Practice and thinking

B. Practice, continued

9. Find $\frac{3}{4} - \frac{1}{3}$.

Answer: _____

10. Simplify the ratio 8 : 14 by dividing both counts by 2.

Answer: _____

11. 3×5

Answer: _____

12. A box is 3 cm by 5 cm by 3 cm. Find its volume.

Answer: _____

C. Think: weekly puzzle

13. Use digits 2, 4, and 8 once each in the form A.BC. Which number is closest to 5?

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Round 748 to the nearest hundred.

Answer: _____

2. Round 7.72 to the nearest whole number.

Answer: _____

3. Use 20×5 to estimate 19.8×5.1.

Answer: _____

4. Use $60 \div 3$ to estimate $59.4 \div 3.1$.

Answer: _____

B. Practice

5. Find $2.1 \div 0.07$.

Answer: _____

6. Find 0.25×0.03.

Answer: _____

7. Find $7 \div 8$.

Answer: _____

8. Find $\frac{2}{4} \div \frac{1}{4}$.

Answer: _____

Practice and thinking

Built-in practice: Fraction operations

9. Find both $4 \div \frac{1}{5}$ and $\frac{1}{5} \div 4$.

Answer: _____

10. Find $4 \times \frac{2}{3}$.

Answer: _____

11. $\frac{2}{3} \times \frac{2}{5}$

Answer: _____

12. $2 \times \frac{2}{5}$

Answer: _____

C. Think

13. A ball costs $4.65 and a card costs $2.78. Find the change from $10.

Answer: _____

14. Lina changes $6.3 \div 0.09$ into $6.3 \div 9$ and gets 0.7. Which change is missing?

Answer: _____

Weekly check

20 minutes. No calculator. Show your work.

1. Find $2.1 \div 0.07$.

Answer: _____

2. Find 0.25×0.03.

Answer: _____

3. Find $7 \div 8$.

Answer: _____

4. Find 0.25×0.04.

Answer: _____

5. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

Answer: _____

Weekly check

6. A machine makes 24 labels in 6 minutes. Find labels per minute and minutes per label.

Answer: _____

7. Jules says $\frac{1}{2} \div \frac{2}{3} = \frac{1}{3}$. Find and fix the error.

Answer: _____

8. Find $\frac{3}{4} - \frac{1}{3}$.

Answer: _____

9. Simplify the ratio 8 : 14 by dividing both counts by 2.

Answer: _____

10. Isaac says $0.6 \times 0.4 = 2.4$ because $6 \times 4 = 24$. Use place value to repair this.

Answer: _____

Recall

Work from memory, then check with your tutor.

R1. 3 markers cost $1.95. Find dollars per marker.

Answer: _____

R2. Find $4 \div \frac{2}{3}$.

Answer: _____

R3. Find 288 ÷ 24.

Answer: _____

R4. 18 shells are 30% of a collection. Find the whole collection.

Answer: _____

S. Return here near the end of the lesson

Distributing works because each group contains __ and __.

Expressions and equivalent forms

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. A sticker sheet costs n dollars. Write the cost of 5 sheets. Find it when n = 2 and when n = 4.

Answer: _____

P2. Find 2a + 5 when a = 6. Write the multiplication before adding.

Answer: _____

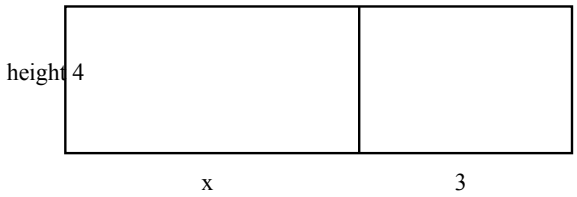
P3. Find 4^2, 4×2, 3^3, and $(\frac{2}{3})^2$. Which pair has different meanings?

Answer: _____

Expressions and equivalent forms

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. A rectangle is 4 units high and x + 3 units wide. Split the width into x and 3. Write its area two ways.



P5. Simplify x + x + x + 2x. Then check your result when x = 3.

Answer: _____

P6. Are 4(t + 2) and 4t + 2 equivalent? Check t = 1. Write a form that is equivalent to 4(t + 2).

Answer: _____

Expressions and equivalent forms

Follow the slide labels. P = problem; F = your turn; D = discussion.

F. Complete: $2(y + 6) = 2y + __$. Check when $y = 5$.

Answer: _____

D. Lena wrote $5(z + 2) = 5z + 2$. Find the mistake and repair it.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Which expression equals 3(2x + 4) for every x? Choose 6x + 12, 6x + 4, or 5x + 7. Show how the 3 acts on both parts.

Answer: _____

Recall

Work from memory, then check with your tutor.

R1. Find 140% of 30 points.

Answer: _____

R2. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

Answer: _____

R3. Find 2.8 ÷ 0.07.

Answer: _____

R4. A snail moves 3 centimeters each minute. Find the distance after 0, 4, and 7 minutes.

Answer: _____

S. Return here near the end of the lesson

I can check a solution by __.

Equations and related quantities

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Solve x + 7 = 19. Show the same change on both sides.

Answer: _____

P2. Four identical bags hold 28 marbles. Write and solve an equation for marbles per bag, b.

Answer: _____

P3. Solve 1.2x = 5.4. Check with multiplication.

Answer: _____

Equations and related quantities

Follow the slide labels. P = problem; F = your turn; D = discussion.

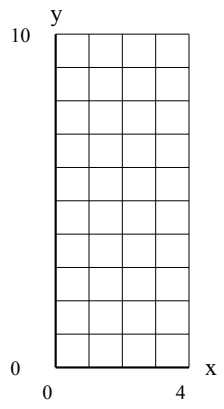
P4. Solve $(\frac{2}{3})x = 10$. Draw three equal parts of x first.

Answer: _____

P5. A game costs $18. Mina has $6.50 saved. Write and solve an equation for the extra amount, a, she needs.

Answer: _____

P6. Notebooks cost $2.50 each. Use c = 2.5n for n = 0, 1, 2, 4. Plot (n, c). Name the input and output.



x: notebooks n; y: cost c

Equations and related quantities

Follow the slide labels. P = problem; F = your turn; D = discussion.

F. Complete: 5x = 35. Divide both sides by __. Then x = __.

Answer: _____

D. Omar solves 6x = 42 by writing x = 42 − 6 = 36. Explain and repair his work.

Answer: _____

Mixed review

M1. 24 shells are 30% of a collection. Find the whole collection.

Answer: _____

M2. Find $2 \div \frac{2}{3}$.

Answer: _____

M3. Find 3.45 + 5.078.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Solve $(\frac{3}{5})x = 12$. Show equal parts or an equation step. Check your value.

Answer: _____

Expressions and equations

The big idea

Expressions name quantities. Equivalent forms describe the same quantity. Equations can be solved by keeping both sides equal.

Words to know

variable: A letter that can stand for a number. Example: In 5n, n can be 2 or 7.

expression: Numbers, variables, and operations without an equals sign. Example: 5n + 3

exponent: The number of equal factors to multiply. Example: 4^2 means 4 × 4.

equivalent: Having the same value for every allowed input. Example: 2(x + 5) and 2x + 10

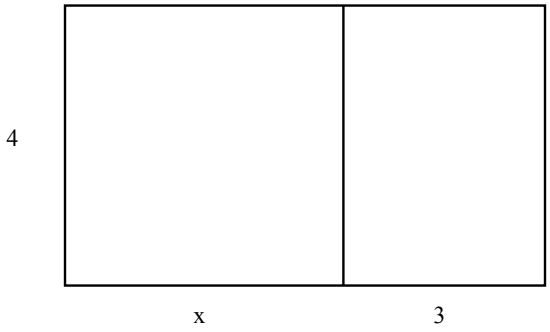
equation: A statement that two quantities are equal. Example: x + 7 = 19

solution: A value that makes an equation true. Example: 12 solves x + 7 = 19.

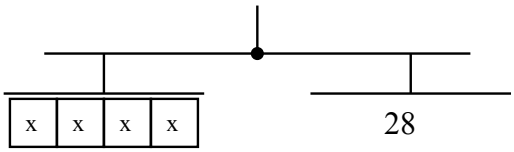
independent variable: The input you choose. Example: Number of notebooks, n.

dependent variable: The output determined by the input. Example: Total cost, c = 2.5n.

How it works



Four rows each contain x + 3. An expression names a quantity. Equivalent expressions name the same quantity for every allowed value.



4 equal bags 28 marbles

Four equal unknown groups balance 28. A solution makes the equation true. The same operation on both sides preserves equality.

Worked examples

1. Find 3(2 + 4).

The parentheses name one group: 2 + 4 = 6.

Three groups give 3 × 6 = 18.

Split each group instead: 3 × 2 + 3 × 4 = 6 + 12 = 18.

Answer: 18

2. Solve x + 2.75 = 8.

Both sides must stay equal.

Subtract 2.75 from each side: x = 5.25.

Check: 5.25 + 2.75 = 8.

Answer: x = 5.25

Common mistakes

Writes $4^2 = 8$. Why: Reads the exponent as multiplication by two. Fix: Write two factors of 4, then compare.

Writes 4(x + 3) = 4x + 3. Why: Loses the four groups of 3. Fix: Shade the numeric part in all four rows.

Checks one value and claims equivalence. Why: A single match can be accidental. Fix: Ask for an area argument that works for any x.

Try it

1. Expand 2(x + 7).

2. Expand 3(x + 6).

3. Solve x + 9 = 23.

4. Solve x + 14 = 31.

5. Solve x + 11 = 18.

Answers

1. 2x + 14 2. 3x + 18 3. x = 14 4. x = 17 5. x = 7

Where this connects

Ratios and fraction division help solve equations. These same equations will describe proportional relationships.

Daily practice

A. Warm-up (3 minutes)

1. 8×3

Answer: _____

2. 8×7

Answer: _____

3. 8×4

Answer: _____

4. 8×9

Answer: _____

B. Practice

5. A rectangle has area $\frac{4}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

Answer: _____

6. Find $\frac{2}{3} + \frac{1}{4}$.

Answer: _____

7. 2×4

Answer: _____

8. Find $2n + 7$ when $n = 5$.

Answer: _____

Practice and thinking

Built-in practice: Expressions and equations

9. Expand $2(x + 7)$.

Answer: _____

10. Find 5^2. Write the repeated factors.

Answer: _____

11. Find $4n + 3$ when $n = 2$.

Answer: _____

12. Each pack has b beads. Write the total in 3 packs. Find it when $b = 4$.

Answer: _____

C. Think

13. Lena wrote $5(z + 2) = 5z + 2$. Find the mistake and repair it.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Write $\frac{1}{2}$ as a decimal and a percent.

Answer: _____

2. Write $\frac{1}{4}$ as a decimal and a percent.

Answer: _____

3. Write $\frac{3}{4}$ as a decimal and a percent.

Answer: _____

4. Write $\frac{1}{8}$ as a decimal and a percent.

Answer: _____

B. Practice

5. Find $\frac{4}{7} \div \frac{2}{5}$. Check the result by multiplying by $\frac{2}{5}$.

Answer: _____

6. 12 shells are 30% of a collection. Find the whole collection.

Answer: _____

7. $36 \div 3$

Answer: _____

8. Solve $(\frac{2}{5})x = 10$.

Answer: _____

Practice and thinking

Built-in practice: Expressions and equations

9. Solve x + 11 = 18.

Answer: _____

10. Solve 7x = 56.

Answer: _____

11. Solve 1.5x = 4.80.

Answer: _____

12. Solve $(\frac{3}{4})x = 12$.

Answer: _____

C. Think

13. Omar solves 6x = 42 by writing x = 42 − 6 = 36. Explain and repair his work.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. 8.2 × 10

Answer: _____

2. 800 ÷ 100

Answer: _____

3. Double 8.5

Answer: _____

4. Half of 48

Answer: _____

B. Practice

5. Solve 1.5x = 4.80.

Answer: _____

6. Expand 4(x + 5).

Answer: _____

7. Find 0.25 × 0.02.

Answer: _____

8. Find 288 ÷ 24.

Answer: _____

Practice and thinking

B. Practice, continued

9. 12 shells are 30% of a collection. Find the whole collection.

Answer: _____

10. Red : blue is 3 : 5. How many blue tiles go with 9 red tiles?

Answer: _____

11. 4 × 6

Answer: _____

12. A box is 4 cm by 6 cm by 3 cm. Find its volume.

Answer: _____

C. Think: weekly puzzle

13. Two whole numbers total 17. One is 5 more than the other. Find both numbers and show a way to check.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Round 848 to the nearest hundred.

Answer: _____

2. Round 8.72 to the nearest whole number.

Answer: _____

3. Use 20 × 5 to estimate 19.8 × 5.1.

Answer: _____

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

Answer: _____

B. Practice

5. Solve 1.5x = 4.80.

Answer: _____

6. Expand 4(x + 5).

Answer: _____

7. Solve 7x = 56.

Answer: _____

8. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

Answer: _____

Practice and thinking

Built-in practice: Decimal practice

9. Find 9.6 ÷ 8.

Answer: _____

10. Find 2.8 ÷ 0.07.

Answer: _____

11. A ball costs $4.65 and a card costs $2.78. Find the change from $10.

Answer: _____

12. Find 10 − 1.286.

Answer: _____

C. Think

13. Each pack has b beads. Write the total in 6 packs. Find it when b = 4.

Answer: _____

14. Omar solves 6x = 42 by writing x = 42 − 6 = 36. Explain and repair his work.

Answer: _____

Weekly check

20 minutes. No calculator. Show your work.

1. Solve $1.5x = 4.80$.

Answer: _____

2. Expand $4(x + 5)$.

Answer: _____

3. Solve $7x = 56$.

Answer: _____

4. Find $3n + 1$ when $n = 4$.

Answer: _____

5. Find $10 - 1.286$.

Answer: _____

Weekly check

6. Find 0.25 × 0.04.

Answer: _____

7. Find 0.25 × 0.02.

Answer: _____

8. 12 shells are 30% of a collection. Find the whole collection.

Answer: _____

9. Red : blue is 3 : 5. How many blue tiles go with 9 red tiles?

Answer: _____

10. Lena wrote 5(z + 2) = 5z + 2. Find the mistake and repair it.

Answer: _____

Recall

Work from memory, then check with your tutor.

R1. Find $2\frac{1}{2} \div \frac{3}{4}$.

Answer: _____

R2. Three notebooks cost $2.45 each. What is the total cost?

Answer: _____

R3. Find 3^2. Write the repeated factors.

Answer: _____

R4. Find $4 \div \frac{2}{3}$.

Answer: _____

S. Return here near the end of the lesson

For this base, the matching height is __ because __.

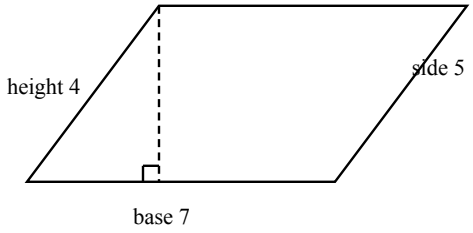
Area from a base and height

Follow the slide labels. P = problem; F = your turn; D = discussion.

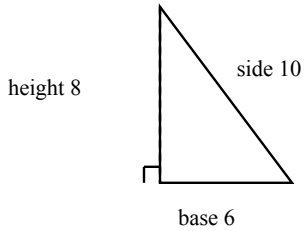
P1. A rectangle is 6 cm long and 2.5 cm wide. Find its area and perimeter.

Answer: _____

P2. A parallelogram has base 7 cm, height 4 cm, and slanted side 5 cm. Cut and move the end triangle. Find its area.



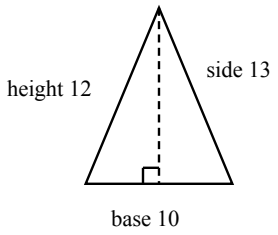
P3. A right triangle has perpendicular legs of 6 cm and 8 cm. Find its area. Use either leg as the base.



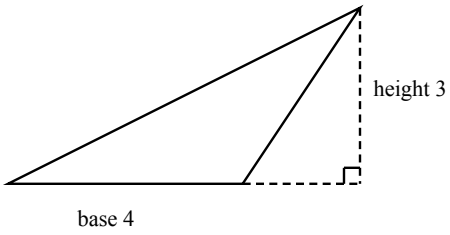
Area from a base and height

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. An isosceles triangle has sides 13, 13, and 10 cm. Its height to the 10 cm base is 12 cm. Find the area.



P5. A triangle has vertices A(0, 0), B(4, 0), and C(6, 3), measured in cm. The height meets the extended base at (6, 0). Find its area.



P6. A triangle has base 5 cm and matching height 2.4 cm. Find its area. Would base × slanted side always work?

Answer: _____

Area from a base and height

Follow the slide labels. P = problem; F = your turn; D = discussion.

F. A triangle has base 8 cm and height 2.5 cm. Complete: $A = \frac{1}{2} \times 8 \times __ = __$.

Answer: _____

D. Theo uses a triangle’s base 9 cm and height 4 cm. He writes $9 \times 4 = 36$ square cm. What is missing?

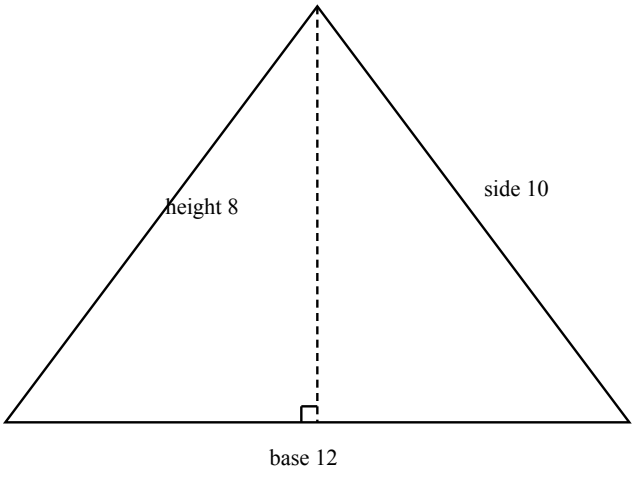
Answer: _____

Exit ticket

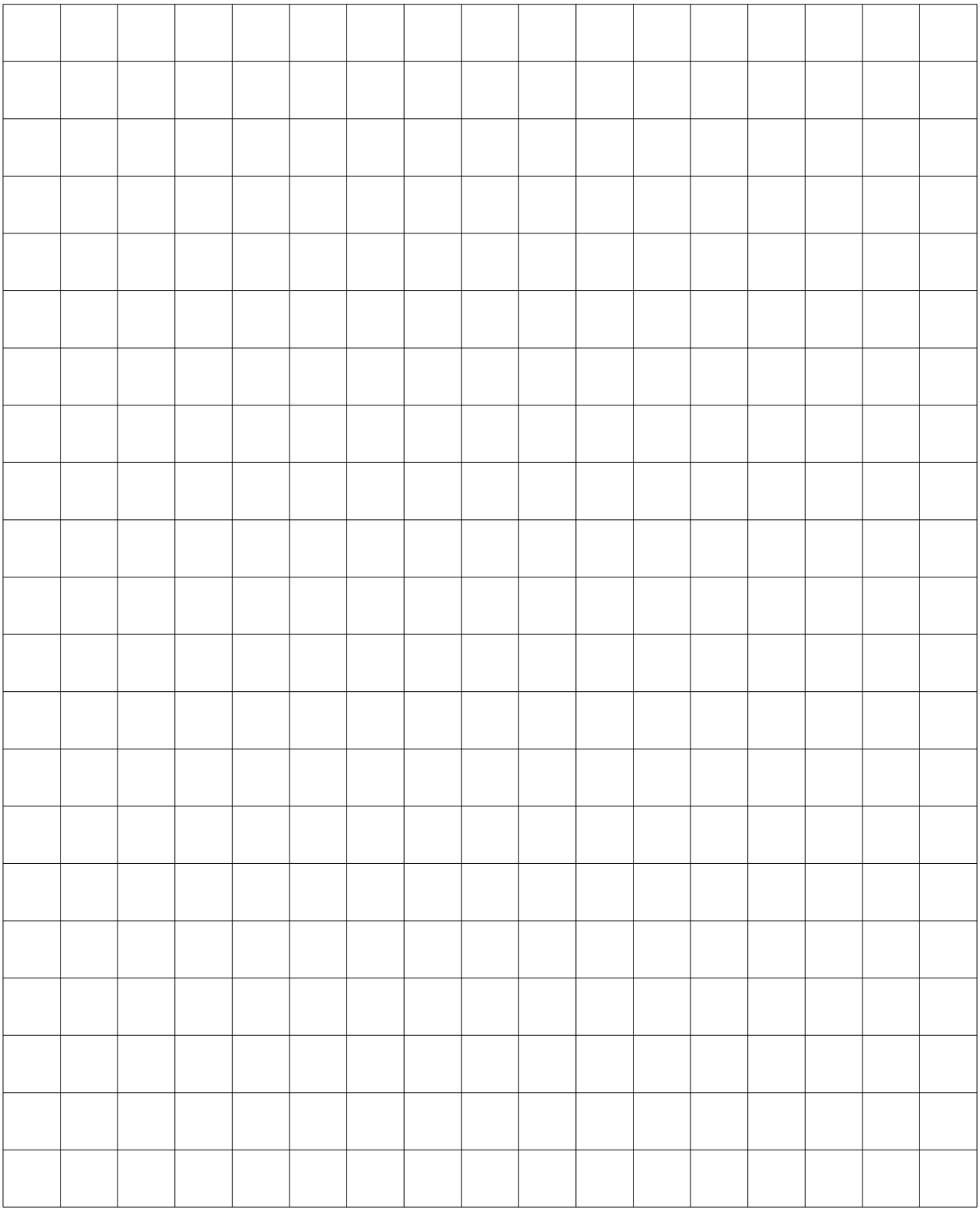
Work on your own. Show a model or reason when the question asks for one.

E. A triangle has sides 10, 10, and 12 cm. The height to the 12 cm base is 8 cm. Find its area and name the length you used as height.

Answer: _____



One Centimeter Grid



Recall

Work from memory, then check with your tutor.

R1. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

Answer: _____

R2. Find $10 - 2.286$.

Answer: _____

R3. Solve $x + 9 = 23$.

Answer: _____

R4. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

Answer: _____

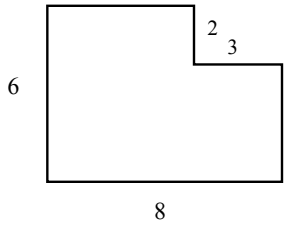
S. Return here near the end of the lesson

To find surface area, I list __ and add __.

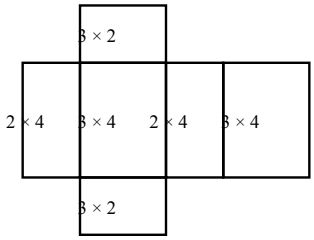
Nets and surface area

Follow the slide labels. P = problem; F = your turn; D = discussion.

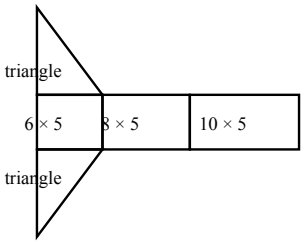
P1. An 8 cm by 6 cm rectangle loses a 3 cm by 2 cm corner from the upper right. Find the L-shaped area two ways.



P2. A closed box is 2 cm by 3 cm by 4 cm. Label all six rectangles on a net. Find surface area and volume.



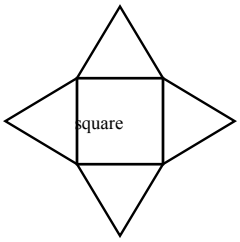
P3. A triangular prism has right-triangle ends with sides 6, 8, and 10 cm. Its length is 5 cm. List the five faces and find surface area.



Nets and surface area

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. A square pyramid has base side 6 cm. Each triangular face has perpendicular height 5 cm. Find its surface area.



P5. A closed 2 cm by 4 cm by 5 cm box has a claimed surface area of 68 square cm. Its 2 by 4 bottom was left out. Repair the answer.

Answer: _____

F. A square pyramid has base side 4 cm and face height 3 cm. Complete: SA = $16 + 4 \times (\frac{1}{2} \times 4 \times __) =$ __.

Answer: _____

Nets and surface area

Follow the slide labels. P = problem; F = your turn; D = discussion.

D. Ava finds a cube’s surface area by adding five faces. Each face is 9 square cm. She gets 45. Explain and repair it.

Answer: _____

Mixed review

M1. A ribbon is $4\ \frac{1}{2}$ meters long. Pieces measure $\frac{3}{8}$ meter. How many pieces fit?

Answer: _____

M2. Find $312 \div 24$.

Answer: _____

M3. Use $y = 2x$ for $x = 1, 3, 5$. Give the ordered pairs.

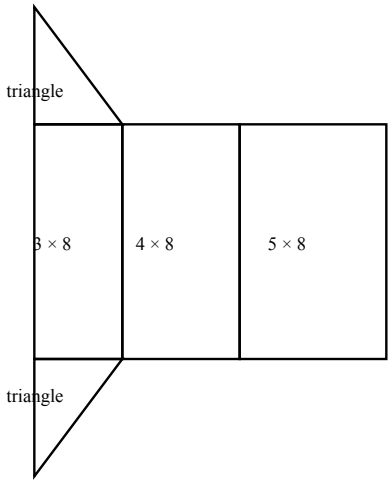
Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. A triangular prism has right-triangle ends with sides 3, 4, and 5 cm. It is 8 cm long. Find the area of all five faces together.

Answer: _____



Area and surface area

The big idea

Area counts square units that cover a surface. Cutting and rearranging reveal triangle and parallelogram formulas. A net helps count every face of a solid.

Words to know

area: The amount of flat surface inside a shape. Example: A 2 by 3 rectangle covers 6 square units.

base: The side chosen to match a height. Example: The bottom side can be the base.

height: The perpendicular distance from the base line to the opposite side or vertex. Example: A height meets the base line at a right angle.

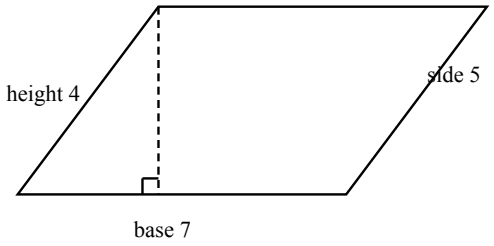
composite polygon: A shape made by joining simpler polygons. Example: An L can split into two rectangles.

net: A flat arrangement that folds into a solid. Example: Six rectangles can form a box.

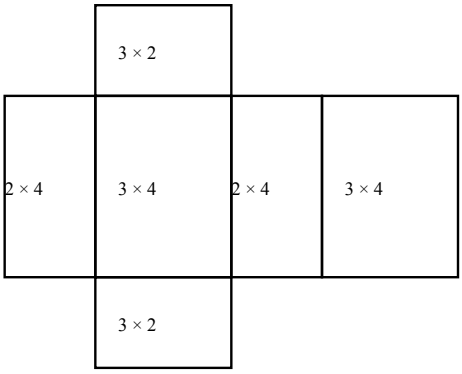
surface area: The total area of all outside faces. Example: A closed box includes its bottom.

volume: The space inside a solid. Example: Volume uses cubic units.

How it works



Cut and slide the left end to make a rectangle. A parallelogram rearranges into a rectangle. Two copies of a triangle make a parallelogram.



Match each rectangle to one face. A net unfolds every face. Surface area adds face areas in square units.

Worked examples

1. A parallelogram has base 6 cm and height 3 cm.

Cut a triangle from one end and move it to the other.

The new rectangle has the same area, base 6 and height 3.

Area = 6 × 3 = 18 square cm.

Answer: 18 square cm

2. Find the surface area of a 2 cm by 2 cm by 3 cm closed box.

Two square ends each have area 4.

Four side rectangles each have area 6.

Total area = 2 × 4 + 4 × 6 = 32 square cm.

Answer: 32 square cm

Common mistakes

Uses a slanted side as height. Why: Treats any given length as interchangeable. Fix: Trace the perpendicular distance with a ruler.

Omits the half for a triangle. Why: Uses the containing parallelogram. Fix: Pair congruent copies and split the area equally.

Uses cm for area. Why: Confuses boundary length and surface coverage. Fix: Count a row of unit squares.

Try it

1. A rectangle is 7 cm by 3 cm. Find its area.

2. A rectangle is 8 cm by 2 cm. Find its area.

3. Find the surface area of a closed 2 by 3 by 5 cm box.

4. Find the surface area of a closed 3 by 3 by 4 cm box.

5. Find the surface area of a closed 2 by 4 by 6 cm box.

Answers

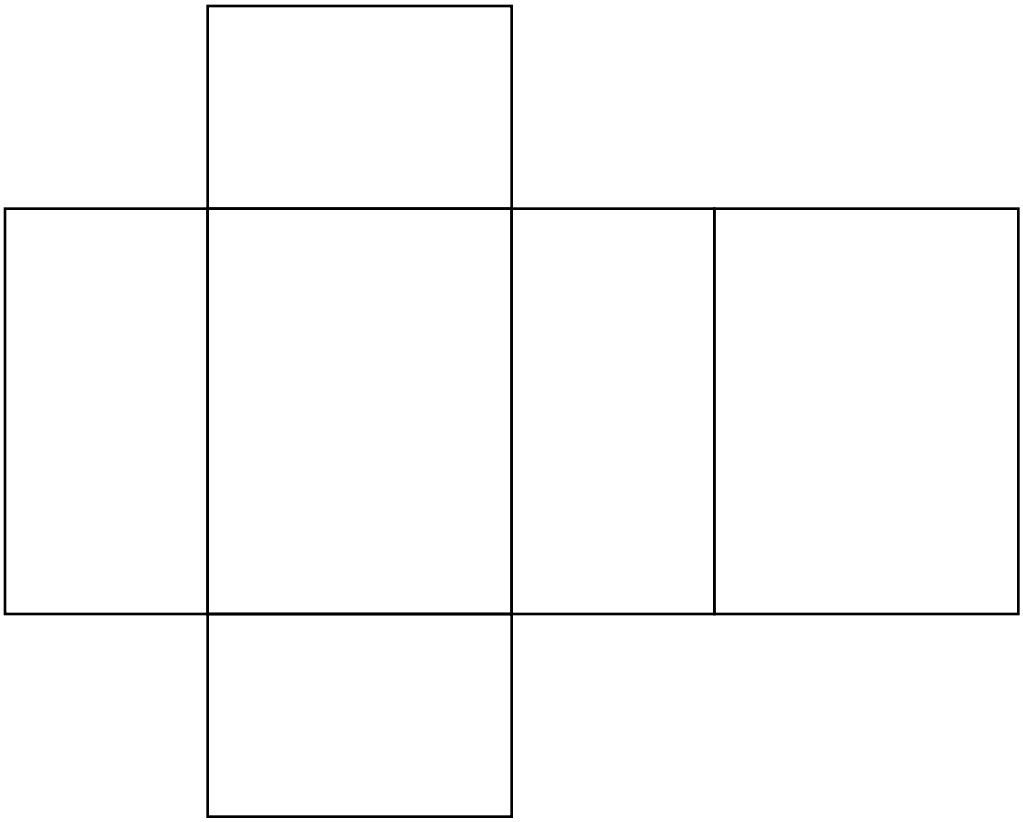
1. 21 square cm 2. 16 square cm 3. 62 square cm 4. 66 square cm 5. 88 square cm

Where this connects

Multiplication and fraction multiplication measure flat regions. Nets connect those regions to three-dimensional solids.

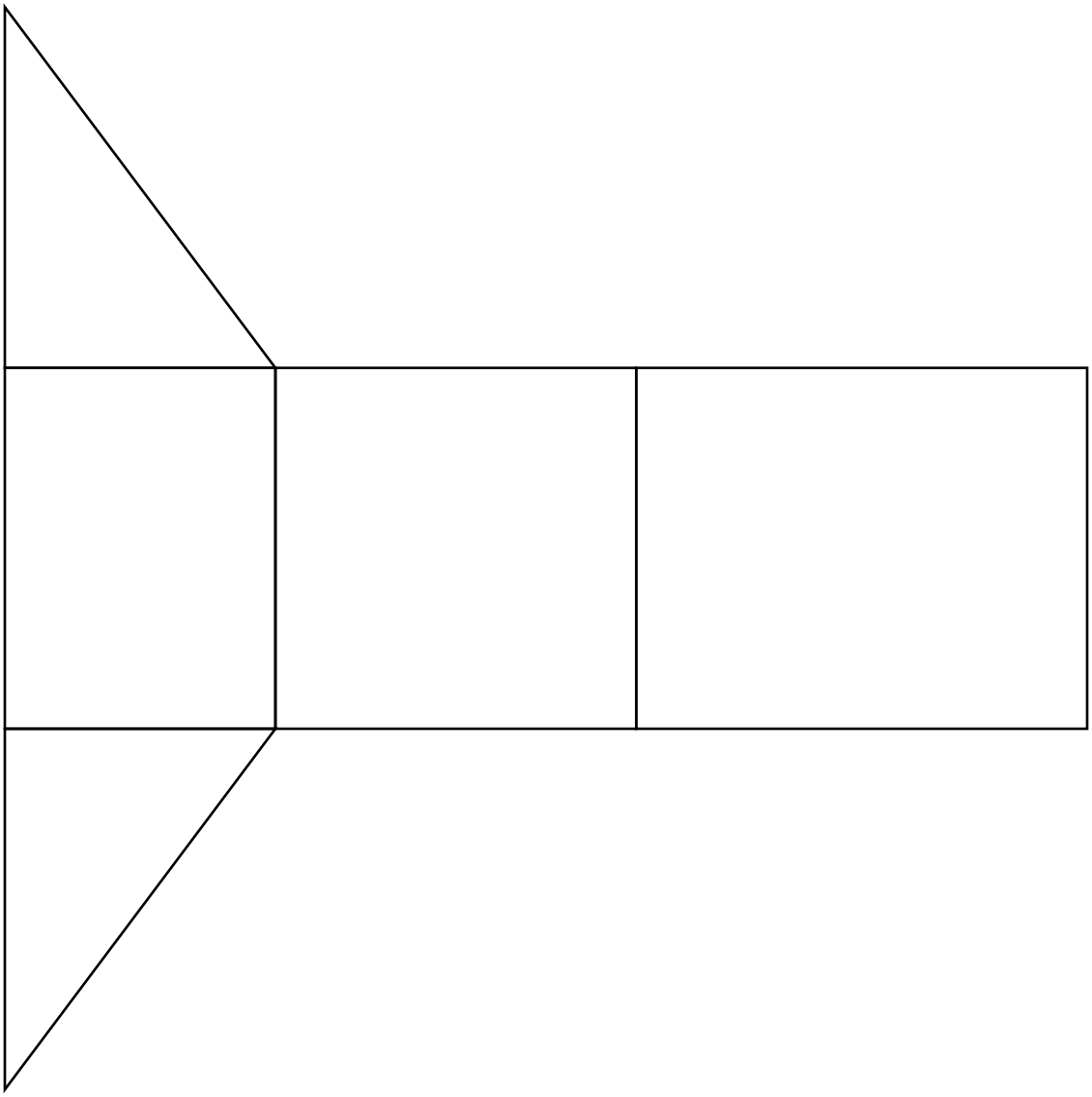
Rectangular prism net

Cut the outside boundary. Fold the internal edges. Tape the faces.



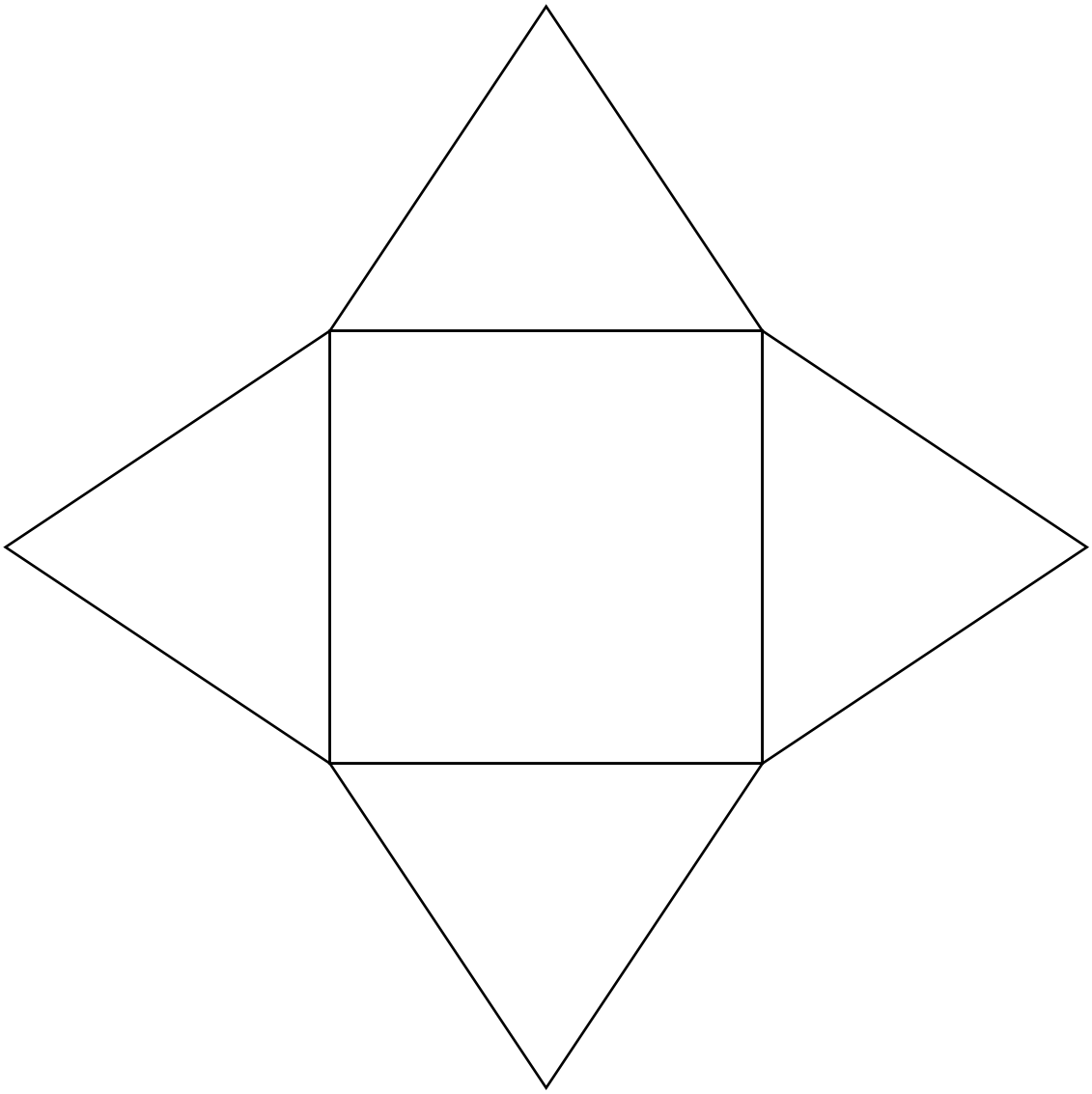
Edges are 2, 3 and 4 units. Shared edges have matching lengths.

Triangular prism net



Triangles: side lengths 3, 4, 5. Prism length: 4 units.

Square pyramid net



Square side: 4 units. Each triangular face has height 3 units.

Daily practice

A. Warm-up (3 minutes)

1. 9 × 3

Answer: _____

2. 9 × 7

Answer: _____

3. 9 × 4

Answer: _____

4. 9 × 9

Answer: _____

B. Practice

5. Find 1,344 ÷ 32. Verify by multiplication.

Answer: _____

6. 9 of 12 tiles are green. What percent are green?

Answer: _____

7. 3 × 5

Answer: _____

8. A parallelogram has base 6, height 8, and slanted side 10 cm. Find its area.

Answer: _____

Practice and thinking

Built-in practice: Area, surface area and volume

9. A rectangle is 6 cm by $\frac{3}{2}$ cm. Find its area.

Answer: _____

10. A parallelogram has base 9, height 4, and slanted side 5 cm. Find its area.

Answer: _____

11. A triangle has base 9 cm and height 4 cm. Find its area.

Answer: _____

12. A 9 by 5 cm rectangle loses a 2 by 3 cm corner. Find the remaining area.

Answer: _____

C. Think

13. Theo uses a triangle's base 9 cm and height 4 cm. He writes $9 \times 4 = 36$ square cm. What is missing?

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Write $\frac{1}{2}$ as a decimal and a percent.

Answer: _____

2. Write $\frac{1}{4}$ as a decimal and a percent.

Answer: _____

3. Write $\frac{3}{4}$ as a decimal and a percent.

Answer: _____

4. Write $\frac{1}{8}$ as a decimal and a percent.

Answer: _____

B. Practice

5. Find 3^2. Write the repeated factors.

Answer: _____

6. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

Answer: _____

7. $48 \div 4$

Answer: _____

8. Find the surface area of a closed 3 by 3 by 4 cm box.

Answer: _____

Practice and thinking

Built-in practice: Area, surface area and volume

9. Find the surface area of a closed 2 by 3 by 5 cm box.

Answer: _____

10. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

Answer: _____

11. A square pyramid has base side 4 cm and triangular-face height 3 cm. Find surface area.

Answer: _____

12. A closed cube has side 2 cm. Jo counts five faces and gets 20 square cm. Repair the total.

Answer: _____

C. Think

13. Ava finds a cube’s surface area by adding five faces. Each face is 9 square cm. She gets 45. Explain and repair it.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. 9.2 × 10

Answer: _____

2. 900 ÷ 100

Answer: _____

3. Double 9.5

Answer: _____

4. Half of 54

Answer: _____

B. Practice

5. A 7 by 6 cm rectangle loses a 2 by 2 cm corner. Find the remaining area.

Answer: _____

6. A parallelogram has base 8, height 3, and slanted side 5 cm. Find its area.

Answer: _____

7. Solve x + 9 = 23.

Answer: _____

8. Find 0.8 × 0.7. Is the product below both factors?

Answer: _____

Practice and thinking

B. Practice, continued

9. Find $\frac{3}{4} \times \frac{4}{5}$.

Answer: _____

10. Order $\frac{2}{5}$, 0.45, and $\frac{1}{2}$ from least to greatest.

Answer: _____

11. 5×2

Answer: _____

12. A box is 5 cm by 2 cm by 3 cm. Find its volume.

Answer: _____

C. Think: weekly puzzle

13. A rectangle has whole-number side lengths, perimeter 24 cm, and area 35 square cm. Find its side lengths.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Round 948 to the nearest hundred.

Answer: _____

2. Round 9.72 to the nearest whole number.

Answer: _____

3. Use 20 × 5 to estimate 19.8 × 5.1.

Answer: _____

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

Answer: _____

B. Practice

5. A 7 by 6 cm rectangle loses a 2 by 2 cm corner. Find the remaining area.

Answer: _____

6. A parallelogram has base 8, height 3, and slanted side 5 cm. Find its area.

Answer: _____

7. Find the surface area of a closed 2 by 3 by 5 cm box.

Answer: _____

8. Each pack has b beads. Write the total in 8 packs. Find it when b = 4.

Answer: _____

Practice and thinking

Built-in practice: Expressions and equations

9. Expand 3(x + 6).

Answer: _____

10. Find 2□. Write the repeated factors.

Answer: _____

11. Solve 6x = 54.

Answer: _____

12. Use y = 4x for x = 0, 2, 3. Give the ordered pairs.

Answer: _____

C. Think

13. A prism has right-triangle ends with sides 3, 4, 5 cm and length 9 cm. Find surface area.

Answer: _____

14. Ava finds a cube’s surface area by adding five faces. Each face is 9 square cm. She gets 45. Explain and repair it.

Answer: _____

Weekly check

20 minutes. No calculator. Show your work.

1. A 7 by 6 cm rectangle loses a 2 by 2 cm corner. Find the remaining area.

Answer: _____

2. A parallelogram has base 8, height 3, and slanted side 5 cm. Find its area.

Answer: _____

3. Find the surface area of a closed 2 by 3 by 5 cm box.

Answer: _____

4. A square pyramid has base side 6 cm and triangular-face height 5 cm. Find surface area.

Answer: _____

5. Find $4n + 3$ when $n = 2$.

Answer: _____

Weekly check

6. Find $3 \div 8$.

Answer: _____

7. Solve $1.5x = 4.80$.

Answer: _____

8. Find $\frac{3}{4} \times \frac{4}{5}$.

Answer: _____

9. Order $\frac{2}{5}$, 0.45, and $\frac{1}{2}$ from least to greatest.

Answer: _____

10. Theo uses a triangle's base 9 cm and height 4 cm. He writes $9 \times 4 = 36$ square cm. What is missing?

Answer: _____

Recall

Work from memory, then check with your tutor.

R1. Find 3.45 + 5.078.

Answer: _____

R2. Solve x + 11 = 18.

Answer: _____

R3. A parallelogram has base 6, height 8, and slanted side 10 cm. Find its area.

Answer: _____

R4. Three notebooks cost $2.45 each. What is the total cost?

Answer: _____

S. Return here near the end of the lesson

The greater number lies __. The greater absolute value means __.

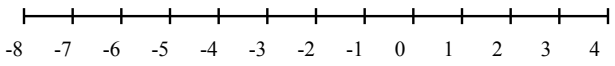
Negative numbers and distance from zero

Follow the slide labels. P = problem; F = your turn; D = discussion.

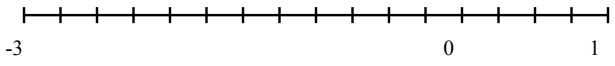
P1. Write a signed number for 4 degrees below zero, 7 m above sea level, and owing $9. State what zero means in each setting.

Answer: _____

P2. Place −6, 1, and −2 on a number line. Which is least?



P3. Place −2.5, $-\frac{3}{4}$, and 0.5 on a number line. Order them from least to greatest.



Negative numbers and distance from zero

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Find the opposite of −9. Then find the opposite of that result. Write −(−9).

Answer: _____

P5. Find |−8|, |8|, and |0|. Describe each as a distance.

Answer: _____

P6. Compare −9 and −4 using < or >. Compare their absolute values. Which temperature is warmer, −9°C or −4°C?

Answer: _____

Negative numbers and distance from zero

Follow the slide labels. P = problem; F = your turn; D = discussion.

F. Complete: −7 is __ units from zero, so |−7| = __. Its opposite is __.

Answer: _____

D. Ben says −10 > −3 because 10 > 3. Use a number line to explain and repair his statement.

Answer: _____

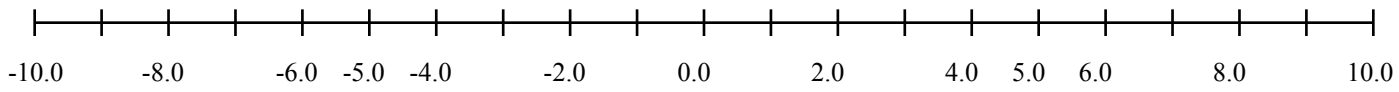
Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Order -6, 1.5, $-\frac{1}{4}$, and -2.5. Which number has the greatest absolute value? Give that absolute value.

Answer: _____

−10 to 10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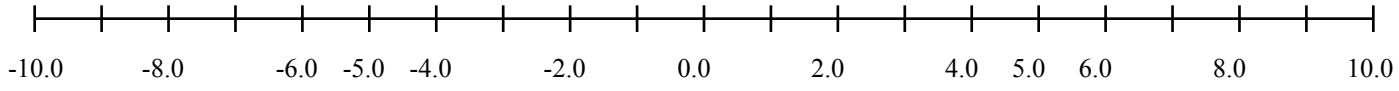
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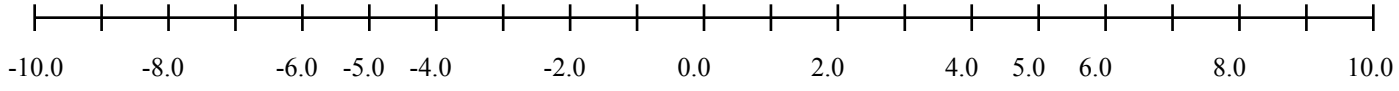
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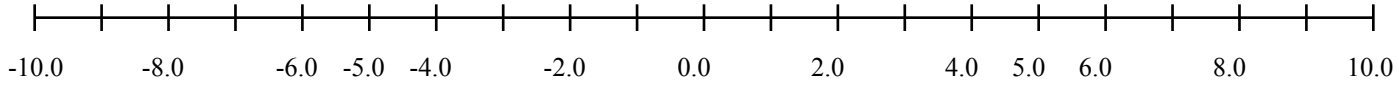
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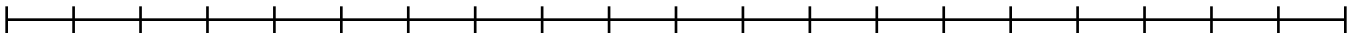
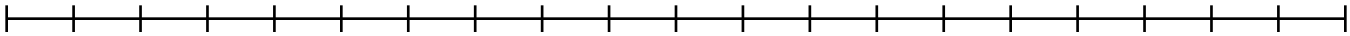
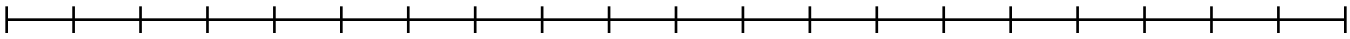
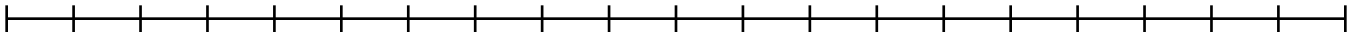
Blank number lines









Recall

Work from memory, then check with your tutor.

R1. Find $1.4 \div 0.07$.

Answer: _____

R2. Solve $(\frac{2}{5})x = 10$.

Answer: _____

R3. Find the surface area of a closed 2 by 3 by 5 cm box.

Answer: _____

R4. Find $10 - 2.286$.

Answer: _____

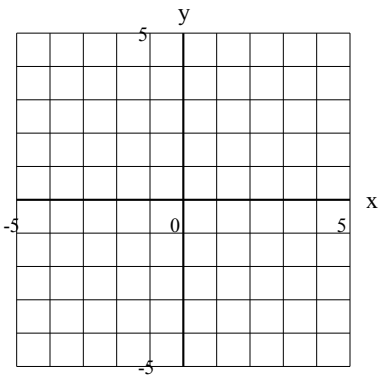
S. Return here near the end of the lesson

A reflection across the y-axis changes __ and keeps __.

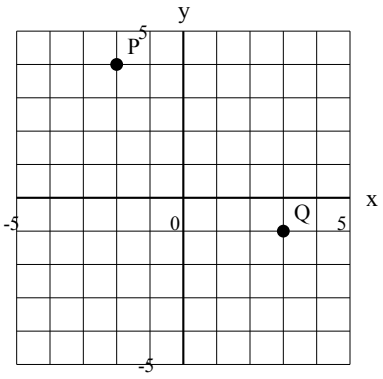
Coordinates and inequalities

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Plot A(2, 3), B(−4, 1), C(−3, −2), and D(1, −4). Name the quadrant of each.



P2. Read P and Q from the grid. State the horizontal move first.



P3. Reflect (−2, 3) across the x-axis. Then reflect the original point across the y-axis.

Answer: _____

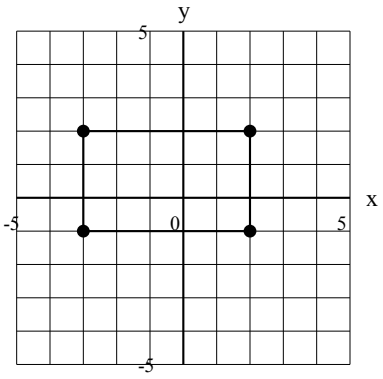
Coordinates and inequalities

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Find the distance from (−2, 3) to (−2, −4), and from (−2, 3) to (5, 3).

Answer: _____

P5. A rectangle has vertices (−3, 2), (2, 2), (2, −1), and (−3, −1). Find its perimeter.



P6. Graph $x > 2$ and $x \geq 2$ on separate number lines. Is 2 a solution of each?

Answer: _____

Coordinates and inequalities

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. A ride requires a height of at least 48 inches. Write an inequality for h. Is 48 included?

Answer: _____

F. Reflect (−5, −2) across the y-axis: (__, __). The distance between the points is __.

Answer: _____

D. Nia writes x > 6 for “at least 6.” She leaves 6 out. Explain and repair her graph.

Answer: _____

Mixed review

M1. Find 1.45 + 3.078.

Answer: _____

M2. Find 3n + 1 when n = 4.

Answer: _____

M3. A rectangle is 8 cm by 2 cm. Find its area.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Reflect A(−4, 1.5) across the y-axis. Give the image and the distance from A to its image.

Answer: _____

Negative numbers, coordinates, and inequalities

The big idea

A signed number gives a position relative to zero. Coordinates use two such positions. Inequalities describe all positions that meet a condition.

Words to know

negative number: A number less than zero. Example: −4 can mean four degrees below zero.

opposites: Numbers the same distance from zero on opposite sides. Example: −6 and 6

absolute value: A number’s distance from zero. Example: $|-6| = 6$

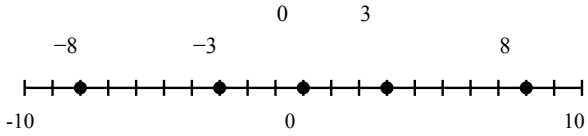
coordinate pair: Two numbers naming horizontal and vertical position. Example: (−2, 3) means left 2, up 3.

reflection: A mirror image across a line. Example: Across the y-axis, (−2, 3) becomes (2, 3).

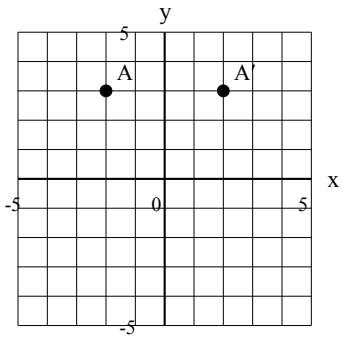
inequality: A comparison using $<$, $>$, $\leq$, or $\geq$. Example: $h \geq 10$ includes 10 and larger numbers.

solution set: All values that make a statement true. Example: $x > 4$ includes 4.5 and 5, but not 4.

How it works



Opposites have equal distances from zero. Numbers increase to the right. Absolute value measures distance from zero, regardless of direction.



The y-axis lies midway between A and A′. A coordinate pair gives horizontal then vertical position. An inequality can describe many solutions.

Worked examples

1. Compare −1.5 and −0.5.

Locate zero, then mark both numbers to its left.

−1.5 lies one unit left of −0.5.

Therefore −1.5 < −0.5, but |−1.5| > |−0.5|.

Answer: −1.5 < −0.5

2. Find the distance from (−1, −3) to (−1, 2).

The x-coordinate stays −1, so the segment is vertical.

From −3 to 0 is 3 units. From 0 to 2 is 2 units.

Total distance = 3 + 2 = 5 units.

Answer: 5 units

Common mistakes

Orders negatives by their digits. Why: Extends positive-number rules unchanged. Fix: Walk rightward on a number line and read the labels.

Gives a negative absolute value. Why: Keeps direction in a distance measure. Fix: Count unit steps from zero.

Says the opposite of −9 is −9. Why: Does not reflect across zero. Fix: Fold a paper number line at zero.

GCF and LCM

The greatest common factor is the largest whole number dividing both numbers. GCF(12, 18) = 6. The least common multiple is the first shared positive multiple. LCM(4, 6) = 12. List factors or multiples to check.

Optional preview

Undo equal steps on both sides. For 2x + 3 = 11, subtract 3 to get 2x = 8. Divide by 2 to get x = 4. A number-line step right adds a positive number.

Try it

1. Order −7, 2, −1 from least to greatest.

2. Order −4, 0, −9 from least to greatest.

3. Reflect (-3, 4) across the y-axis.

4. Reflect (2, -5) across the x-axis.

5. Reflect (-4, -2) across the y-axis.

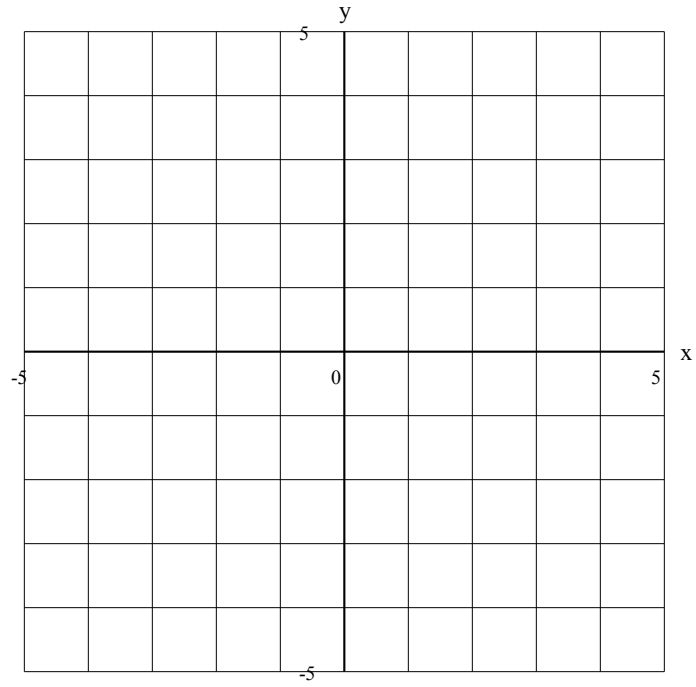
Answers

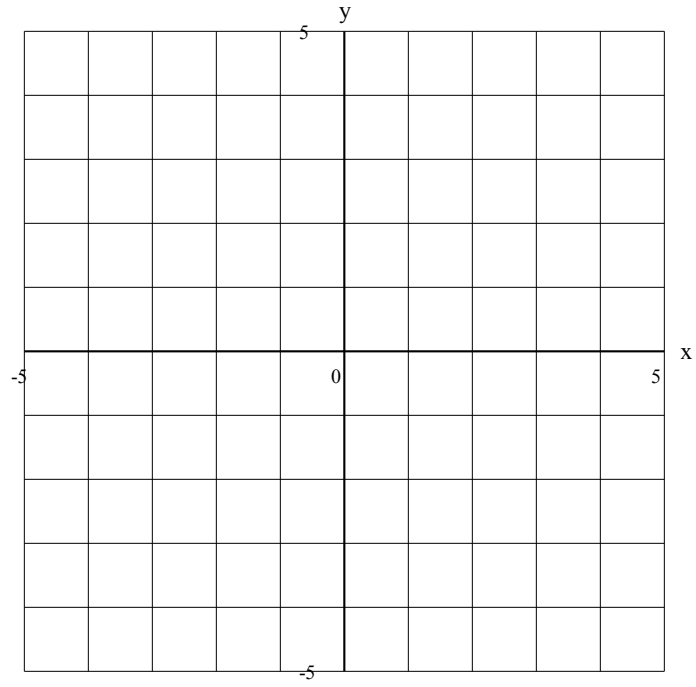
1. −7, −1, 2 2. −9, −4, 0 3. (3, 4) 4. (2, 5) 5. (4, −2)

Where this connects

Number lines extend fraction and decimal ideas to negatives. Coordinates will show proportional relationships and data.

Coordinate planes





Daily practice

A. Warm-up (3 minutes)

1. 10 × 3

Answer: _____

2. 10 × 7

Answer: _____

3. 10 × 4

Answer: _____

4. 10 × 9

Answer: _____

B. Practice

5. A parallelogram has base 6, height 8, and slanted side 10 cm. Find its area.

Answer: _____

6. 9 of 12 tiles are green. What percent are green?

Answer: _____

7. 4 × 6

Answer: _____

8. Find |5| and describe it as a distance.

Answer: _____

Practice and thinking

Built-in practice: Number lines and coordinates

9. Order −4, 0, −9 from least to greatest.

Answer: _____

10. Find |−12| and describe it as a distance.

Answer: _____

11. Find |0| and describe it as a distance.

Answer: _____

12. Find the opposite of −6.

Answer: _____

C. Think

13. Ben says −10 > −3 because 10 > 3. Use a number line to explain and repair his statement.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Write $\frac{1}{2}$ as a decimal and a percent.

Answer: _____

2. Write $\frac{1}{4}$ as a decimal and a percent.

Answer: _____

3. Write $\frac{3}{4}$ as a decimal and a percent.

Answer: _____

4. Write $\frac{1}{8}$ as a decimal and a percent.

Answer: _____

B. Practice

5. Find the surface area of a closed 2 by 3 by 5 cm box.

Answer: _____

6. Five granola bars are shared equally among six children. How much does each get?

Answer: _____

7. $60 \div 5$

Answer: _____

8. Find the distance from (1, -5) to (1, 4).

Answer: _____

Practice and thinking

Built-in practice: Number lines and coordinates

9. Reflect (-3, 4) across the y-axis.

Answer: _____

10. Find the distance from (-4, 2) to (3, 2).

Answer: _____

11. Write an inequality for x that means at least 9. Is the boundary included?

Answer: _____

12. Write an inequality for x that means less than 5. Is the boundary included?

Answer: _____

C. Think

13. Nia writes x > 6 for “at least 6.” She leaves 6 out. Explain and repair her graph.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. 10.2 × 10

Answer: _____

2. 1000 ÷ 100

Answer: _____

3. Double 10.5

Answer: _____

4. Half of 60

Answer: _____

B. Practice

5. Find |0| and describe it as a distance.

Answer: _____

6. Find the opposite of 9.

Answer: _____

7. A parallelogram has base 8, height 3, and slanted side 5 cm. Find its area.

Answer: _____

8. A 10 by 4 cm rectangle loses a 3 by 1 cm corner. Find the remaining area.

Answer: _____

Practice and thinking

B. Practice, continued

9. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

Answer: _____

10. Find 25% of 24 stickers.

Answer: _____

11. List common factors of 12 and 18. Find the greatest common factor.

Answer: _____

12. List common factors of 8 and 20. Find the greatest common factor.

Answer: _____

C. Think: weekly puzzle

13. A point lies left of the y-axis and above the x-axis. It is 3 units from the y-axis and 2 from the x-axis. Find it, then reflect it across both axes in turn.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Round 1048 to the nearest hundred.

Answer: _____

2. Round 10.72 to the nearest whole number.

Answer: _____

3. Use 20×5 to estimate 19.8×5.1.

Answer: _____

4. Use $60 \div 3$ to estimate $59.4 \div 3.1$.

Answer: _____

B. Practice

5. Find $|0|$ and describe it as a distance.

Answer: _____

6. Find the opposite of 9.

Answer: _____

7. Reflect (-3, 4) across the y-axis.

Answer: _____

8. A triangle has base 9 cm and height 4 cm. Find its area.

Answer: _____

Practice and thinking

Built-in practice: Area, surface area and volume

9. A rectangle is 7 cm by 3 cm. Find its area.

Answer: _____

10. A 7 by 6 cm rectangle loses a 2 by 2 cm corner. Find the remaining area.

Answer: _____

11. Find the surface area of a closed 3 by 3 by 4 cm box.

Answer: _____

12. A box has edges $\frac{1}{2}$, $\frac{2}{3}$, and 2 meters. Find the volume.

Answer: _____

C. Think

13. Write an inequality for x that means at least 9. Is the boundary included?

Answer: _____

14. Nia writes x > 6 for “at least 6.” She leaves 6 out. Explain and repair her graph.

Answer: _____

Stretch (optional)

15. Start at −3. Move 5 units right. Where do you land?

Answer: _____

16. Solve 2x + 3 = 11.

Answer: _____

Weekly check

20 minutes. No calculator. Show your work.

1. Find |0| and describe it as a distance.

Answer: _____

2. Find the opposite of 9.

Answer: _____

3. Reflect (-3, 4) across the y-axis.

Answer: _____

4. Find |−12| and describe it as a distance.

Answer: _____

5. Solve 6x = 54.

Answer: _____

Weekly check

6. Solve $1.5x = 4.80$.

Answer: _____

7. A square pyramid has base side 4 cm and triangular-face height 3 cm. Find surface area.

Answer: _____

8. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

Answer: _____

9. Find 25% of 24 stickers.

Answer: _____

10. Ben says $-10 > -3$ because $10 > 3$. Use a number line to explain and repair his statement.

Answer: _____

Recall

Work from memory, then check with your tutor.

R1. Use y = 2x for x = 1, 3, 5. Give the ordered pairs.

Answer: _____

R2. Find the surface area of a closed 2 by 4 by 6 cm box.

Answer: _____

R3. Find the distance from (-2, -3) to (-2, 2).

Answer: _____

R4. Solve x + 11 = 18.

Answer: _____

S. Return here near the end of the lesson

The mean tells __. The median tells __.

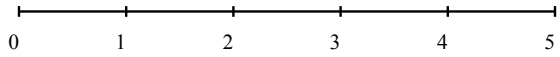
Data displays and center

Follow the slide labels. P = problem; F = your turn; D = discussion.

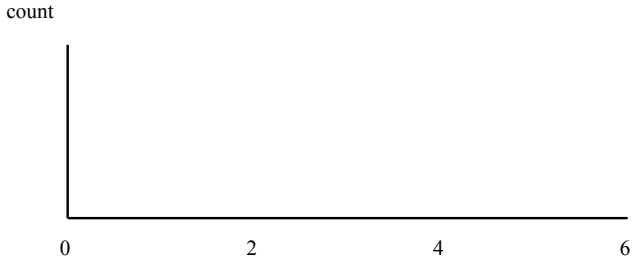
P1. Which asks a statistical question: “How many pages are in this book?” or “How many pages are in each library book on this shelf?” Explain.

Answer: _____

P2. Build a dot plot for 1, 2, 2, 3, 4, 4, 5 books read. Describe its shape and total number of dots.



P3. Build a histogram for 0, 1, 1, 2, 2, 3, 4, 5. Use bins 0 to under 2, 2 to under 4, and 4 to under 6.



Data displays and center

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Four jars hold 2, 4, 5, and 9 counters. Share all counters equally. Find the mean and show how it balances the deviations.

Answer: _____

P5. Find each median: (a) 8, 2, 4, 4, 7; (b) 5, 2, 9, 3, 6, 5. Order first.

Answer: _____

P6. Data are 2, 3, 4, 4, 7. Replace 7 with 27. Compare the old and new mean and median. Which center changes less?

Answer: _____

Data displays and center

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. Four game scores have mean 6. Three scores are 3, 5, and 7. Find the missing score.

Answer: _____

F. Ordered data are 1, 3, 6, 8. Complete: median = (3 + __) ÷ 2 = __.

Answer: _____

D. Luis finds the median of 9, 2, 6, 3 by averaging 2 and 6. He gets 4. What should he change?

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Data are 8, 2, 11, 5, 6, 4. Find the median and the mean. Show your ordered list.

Answer: _____

Spread and box plots

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Data are 3, 7, 8, 10, 12. Find the range.

Answer: _____

P2. Find the five-number summary of 1, 3, 4, 6, 8, 9, 11. Exclude the median from each half.

Answer: _____

P3. Find the five-number summary and IQR of 1, 2, 4, 6, 8, 10, 12, 13.

Answer: _____

Spread and box plots

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Data are 2, 4, 6, 8. Find the mean, each distance from the mean, and the MAD.

Answer: _____

P5. A box plot has five-number summary 0, 2.5, 5, 9, 12. Which interval represents the upper quarter? Does a longer interval mean more data?

Answer: _____

P6. Compare A: 4, 4, 6, 6 and B: 1, 3, 7, 9. Find each mean and MAD. Which set is more spread out?

Answer: _____

Spread and box plots

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. Data are 1, 3, 4, 5, 6, 7, 9, 11. Replace 11 with 31. Compare the old and new range and IQR.

Answer: _____

F. Data are 3, 5, 7, 9. Mean = 6. Distances are 3, __, __, 3. MAD = __.

Answer: _____

D. Zoe averages −3, −1, 1, and 3 to find MAD. She gets 0. What should she average?

Answer: _____

Spread and box plots

Follow the slide labels. P = problem; F = your turn; D = discussion.

S. IQR measures __. MAD measures __.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. For 1, 3, 3, 5, 7, 9, 11, 13, find Q1, the median, Q3, and the IQR. Show the two halves.

Answer: _____

Grade 6 Cumulative Check

30 minutes. No calculator. Show your work.

1. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

Answer: _____

2. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

Answer: _____

3. A bus covers 10 miles in 25 minutes at constant speed. How many miles in one hour?

Answer: _____

4. Find 140% of 30 points.

Answer: _____

5. 12 shells are 30% of a collection. Find the whole collection.

Answer: _____

6. How many $\frac{3}{4}$-liter groups fit in $\frac{7}{8}$ liter? Include part of a group.

Answer: _____

Grade 6 Cumulative Check

7. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

Answer: _____

8. Which is the reciprocal of $2\ \frac{1}{2}$: $\frac{2}{5}$ or $\frac{5}{2}$? Explain using a product of 1.

Answer: _____

9. Three notebooks cost $2.45 each. What is the total cost?

Answer: _____

10. Find $9.6 \div 8$.

Answer: _____

11. Find $1{,}344 \div 32$. Verify by multiplication.

Answer: _____

12. Expand $2(x + 7)$.

Answer: _____

Grade 6 Cumulative Check

13. Solve $x + 9 = 23$.

Answer: _____

14. Solve $(\frac{3}{4})x = 12$.

Answer: _____

15. A parallelogram has base 9, height 4, and slanted side 5 cm. Find its area.

Answer: _____

16. A triangle has base 9 cm and height 4 cm. Find its area.

Answer: _____

17. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

Answer: _____

18. Order -7, 2, -1 from least to greatest.

Answer: _____

Grade 6 Cumulative Check

19. Reflect (-3, 4) across the y-axis.

Answer: _____

20. Write an inequality for x that means at least 9. Is the boundary included?

Answer: _____

21. Find the mean of 2, 3, 7, 8.

Answer: _____

22. Find the median of 9, 1, 5, 3, 7.

Answer: _____

23. Find Q1, Q3, and IQR for 1, 2, 4, 5, 7, 8, 10, 11. Exclude the median if the count is odd.

Answer: _____

24. Find the MAD of 1, 3, 5, 7.

Answer: _____

Data, center, and spread

The big idea

Data vary. Center describes a typical value, and spread describes the variation. A display makes both easier to see.

Words to know

statistical question: A question answered with data that are expected to vary. Example: How many books did each classmate read?

mean: The total shared equally among all data values. Example: The mean of 2 and 6 is 4.

median: The middle of ordered data. With two middles, use their mean. Example: The median of 1, 4, 8 is 4.

histogram: A display counting values in number intervals. Example: A bar can count heights from 120 to under 130 cm.

range: Largest value minus smallest value. Example: For 2, 4, 9, range is 7.

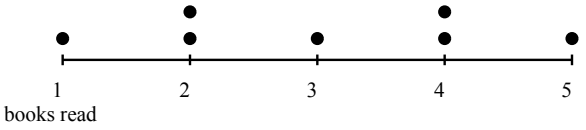
quartiles: Medians of the lower and upper halves of ordered data. Example: Exclude the middle value when the count is odd.

IQR: Upper quartile minus lower quartile. Example: Q3 = 9 and Q1 = 3 give IQR 6.

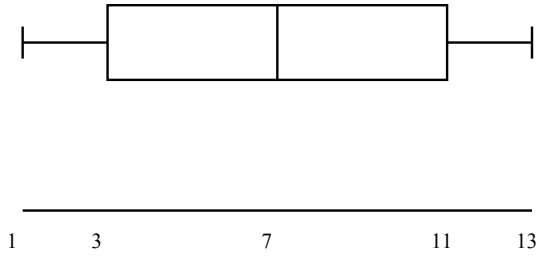
MAD: The mean of the distances from the mean. Example: Use positive distances before averaging.

How it works

The seven values balance around 3. A statistical question expects different answers. Mean shares the total equally; median locates the ordered middle.



The median splits the ordered data. Quartiles split each half. Center describes a typical value. Spread describes how far values are apart or away from the center.



Worked examples

1. Find the mean of 3, 4, 5, and 8.

The total is 3 + 4 + 5 + 8 = 20.

There are four values, so share 20 among four groups.

Mean = 20 ÷ 4 = 5. The deviations −2, −1, 0, 3 balance to zero.

Answer: 5

2. Find the IQR for 2, 4, 6, 8, 10, 12.

Split the ordered data into 2, 4, 6 and 8, 10, 12.

The two half-medians are Q1 = 4 and Q3 = 10.

IQR = 10 − 4 = 6.

Answer: 6

Common mistakes

Omits repeated values in a dot plot. Why: Treats the display as a list of categories. Fix: Cross out each raw value after plotting one dot.

Chooses one middle value in an even set. Why: Uses an odd-set rule. Fix: Mark both middle values and find their midpoint.

Uses median for every total-sharing question. Why: Does not connect mean to a fixed total. Fix: Redistribute real counters equally.

Optional preview

A 25% discount leaves 75% of the starting price. For 3(x + 2) = 18, divide both sides by 3, then subtract 2. This gives x = 4.

Try it

1. Find the mean of 2, 3, 7, 8.

2. Find the mean of 1, 4, 5, 6.

3. Find the range of 2, 5, 8, 11.

4. Find the range of 3, 4, 6, 9, 15.

5. Find the range of 0, 2, 5, 10.

Answers

1. 5 2. 4 3. 9 4. 12 5. 10

Where this connects

Mean uses equal sharing, and MAD uses distance from zero. A cumulative check links these new ideas with earlier Grade 6 work.

Daily practice

A. Warm-up (3 minutes)

1. 11 × 3

Answer: _____

2. 11 × 7

Answer: _____

3. 11 × 4

Answer: _____

4. 11 × 9

Answer: _____

B. Practice

5. Reflect (-3, 4) across the y-axis.

Answer: _____

6. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

Answer: _____

7. 5 × 2

Answer: _____

8. Find the mean of 1, 4, 5, 6.

Answer: _____

Practice and thinking

Built-in practice: Data and statistics

9. Find the mean of 2, 3, 7, 8.

Answer: _____

10. Find the median of 9, 1, 5, 3, 7.

Answer: _____

11. Find the median of 8, 2, 4, 6.

Answer: _____

12. Four scores have mean 5. Three are 2, 4, 7. Find the fourth.

Answer: _____

C. Think

13. Luis finds the median of 9, 2, 6, 3 by averaging 2 and 6. He gets 4. What should he change?

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Write $\frac{1}{2}$ as a decimal and a percent.

Answer: _____

2. Write $\frac{1}{4}$ as a decimal and a percent.

Answer: _____

3. Write $\frac{3}{4}$ as a decimal and a percent.

Answer: _____

4. Write $\frac{1}{8}$ as a decimal and a percent.

Answer: _____

B. Practice

5. Find $|-12|$ and describe it as a distance.

Answer: _____

6. Find $2.1 \div 0.07$.

Answer: _____

7. $72 \div 6$

Answer: _____

8. Find the MAD of 3, 3, 9, 9.

Answer: _____

Practice and thinking

Built-in practice: Data and statistics

9. Find the range of 2, 5, 8, 11.

Answer: _____

10. Find Q1, Q3, and IQR for 1, 2, 4, 5, 7, 8, 10, 11. Exclude the median if the count is odd.

Answer: _____

11. Find the MAD of 1, 3, 5, 7.

Answer: _____

12. A: 3, 4, 5. B: 1, 4, 7. Compare their means and ranges.

Answer: _____

C. Think

13. Zoe averages −3, −1, 1, and 3 to find MAD. She gets 0. What should she average?

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. 11.2 × 10

Answer: _____

2. 1100 ÷ 100

Answer: _____

3. Double 11.5

Answer: _____

4. Half of 66

Answer: _____

B. Practice

5. Find the range of 0, 2, 5, 10.

Answer: _____

6. Find the MAD of 2, 5, 5, 8.

Answer: _____

7. Find |−3| and describe it as a distance.

Answer: _____

8. A prism has right-triangle ends with sides 3, 4, 5 cm and length 6 cm. Find surface area.

Answer: _____

Practice and thinking

B. Practice, continued

9. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

Answer: _____

10. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

Answer: _____

11. 7×4

Answer: _____

12. A box is 7 cm by 4 cm by 3 cm. Find its volume.

Answer: _____

C. Think: weekly puzzle

13. Find two sets of five different whole numbers with minimum 2, maximum 12, median 5, and mean 6.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Round 1148 to the nearest hundred.

Answer: _____

2. Round 11.72 to the nearest whole number.

Answer: _____

3. Use 20×5 to estimate 19.8×5.1.

Answer: _____

4. Use $60 \div 3$ to estimate $59.4 \div 3.1$.

Answer: _____

B. Practice

5. Find the range of 0, 2, 5, 10.

Answer: _____

6. Find the MAD of 2, 5, 5, 8.

Answer: _____

7. Data are 1, 2, 4, 4, 5, 6, 6. How many dots belong above 4 in a dot plot?

Answer: _____

8. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

Answer: _____

Practice and thinking

Built-in practice: Number lines and coordinates

9. Order −3, 1, −6 from least to greatest.

Answer: _____

10. Find |5| and describe it as a distance.

Answer: _____

11. Reflect (2, -5) across the x-axis.

Answer: _____

12. Find the distance from (-2, -3) to (-2, 2).

Answer: _____

C. Think

13. Find Q1, Q3, and IQR for 0, 2, 3, 5, 7, 9, 10, 12. Exclude the median if the count is odd.

Answer: _____

14. Zoe averages −3, −1, 1, and 3 to find MAD. She gets 0. What should she average?

Answer: _____

Stretch (optional)

15. A $40 item is 25% off. Find the sale price.

Answer: _____

16. Solve 3(x + 2) = 18.

Answer: _____

Weekly check

20 minutes. No calculator. Show your work.

1. Find the range of 0, 2, 5, 10.

Answer: _____

2. Find the MAD of 2, 5, 5, 8.

Answer: _____

3. Data are 1, 2, 4, 4, 5, 6, 6. How many dots belong above 4 in a dot plot?

Answer: _____

4. Four scores have mean 7. Three are 4, 5, 9. Find the fourth.

Answer: _____

5. A triangle has base 9 cm and height 4 cm. Find its area.

Answer: _____

Weekly check

6. Find the distance from (-4, 2) to (3, 2).

Answer: _____

7. Write an inequality for x that means no more than 12. Is the boundary included?

Answer: _____

8. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

Answer: _____

9. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

Answer: _____

10. Luis finds the median of 9, 2, 6, 3 by averaging 2 and 6. He gets 4. What should he change?

Answer: _____

Recall

Work from memory, then check with your tutor.

R1. A rectangle is 8 cm by 2 cm. Find its area.

Answer: _____

R2. List multiples of 4 and 6. Find the least positive common multiple.

Answer: _____

R3. Data are 1, 2, 4, 4, 5, 6, 6. How many dots belong above 4 in a dot plot?

Answer: _____

R4. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

Answer: _____

S. Return here near the end of the lesson

A proportional relationship has __ for every nonzero input and passes through __.

Scale and proportional relationships

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. A 4 cm by 6 cm rectangle is copied with scale factor 2. Find the new side lengths and compare areas.

Answer: _____

P2. A triangle has side lengths 6, 8, and 10 cm. Copy it with scale factor $\frac{1}{2}$. Find its side lengths. What happens to its angles?

Answer: _____

P3. A map uses 1 cm for 25 km. Two towns are 3.6 cm apart on the map. Find their actual distance.

Answer: _____

Scale and proportional relationships

Follow the slide labels. P = problem; F = your turn; D = discussion.

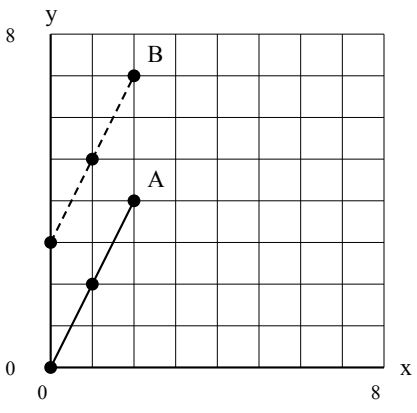
P4. A drawing has area 5 square cm. A scaled copy has all lengths three times as large. Find its area.

Answer: _____

P5. A table pairs x = 1, 2, 4 with y = 2.5, 5, 10. Find k and write y = kx.

Answer: _____

P6. Compare two straight-line graphs: A passes through (0, 0), (1, 2), (2, 4). B passes through (0, 3), (1, 5), (2, 7). Which is proportional?



Scale and proportional relationships

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. A ride costs $4 to start and $3 per mile. Write cost c for m miles. Is it proportional?

Answer: _____

P8. A faucet fills $\frac{3}{4}$ liter in $\frac{1}{2}$ minute at a constant rate. Write liters L after t minutes.

Answer: _____

F. A proportional relationship has (x, y) = (4, 14). Complete: k = 14 ÷ __ = __, so y = __x.

Answer: _____

Scale and proportional relationships

Follow the slide labels. P = problem; F = your turn; D = discussion.

D. Eli triples every length of a drawing. He says its area also triples. Explain and repair his claim.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

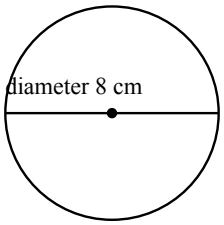
E. A table pairs x = 1, 3, 5 with y = 5, 11, 17. Is y proportional to x? Support your answer with ratios or the value at zero.

Answer: _____

Circles and first previews

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. A circle has diameter 8 cm. Find its circumference in terms of π and using 3.14.



P2. A circle has radius 3 cm. Find its diameter and circumference in terms of π.

Answer: _____

P3. Find the area of a circle with radius 4 cm. Give an exact answer and an approximation using 3.14.

Answer: _____

Circles and first previews

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. A circle has diameter 6 cm. Find its area in terms of π and using 3.14.

Answer: _____

P5. Circle A has diameter 4 cm and circumference 4π cm. Circle B has diameter 10 cm and circumference 10π cm. Compare C ÷ d.

Answer: _____

P6. A collection grows from 80 cards to 100. Find the percent increase. Then find the percent decrease from 100 back to 80.

Answer: _____

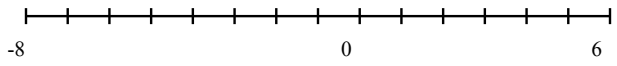
Circles and first previews

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. Start with 80 tokens. Increase by 25%, then decrease the result by 25%. Find the final amount. Explain why it is not 80.

Answer: _____

P8. Use steps on a number line for −2 + 5, −3 + (−4), and 2 − 6.



F. A circle has diameter 16 cm. Complete: radius = __; area = π × __² = __π square cm.

Answer: _____

Circles and first previews

Follow the slide labels. P = problem; F = your turn; D = discussion.

D. Maya finds area from a diameter of 14 cm by writing $\pi \times 14^2 = 196\pi$. Explain and repair it.

Answer: _____

S. For circle area, I first identify __. A percent change compares the change with __.

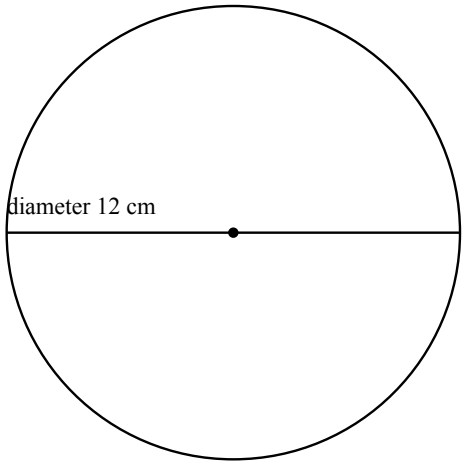
Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. A circle has diameter 12 cm. Find its area in terms of π and using 3.14. Show the radius you used.

Answer: _____



Final Mixed Review

15 minutes. No calculator. Show your work.

1. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

Answer: _____

2. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

Answer: _____

3. A machine makes 24 labels in 6 minutes. Find labels per minute and minutes per label.

Answer: _____

4. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

Answer: _____

5. Find $9 \div 40$.

Answer: _____

Final Mixed Review

6. Expand $2(x + 7)$.

Answer: _____

7. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

Answer: _____

8. Find the opposite of -6.

Answer: _____

9. Find Q1, Q3, and IQR for 0, 2, 3, 5, 7, 9, 10, 12. Exclude the median if the count is odd.

Answer: _____

10. A circle has diameter 3 cm. Find circumference exactly and using 3.14.

Answer: _____

Scale, proportions, circles, and review

The big idea

A fixed multiplier connects scaled lengths and proportional quantities. Circle formulas use radius or diameter in different ways. The final weekdays revisit the entire program.

Words to know

scale factor: The multiplier from an original length to a copy length. Example: A factor of 2 doubles every length.

scaled copy: A shape with matching angles and one multiplier for every length. Example: A 2 by 3 rectangle becomes 4 by 6.

constant of proportionality: The same y per one x in a proportional relationship. Example: For $y = 4x$, $k = 4$.

proportional: Related by a fixed multiplier with zero matched to zero. Example: At $3 each, cost = 3 × item count.

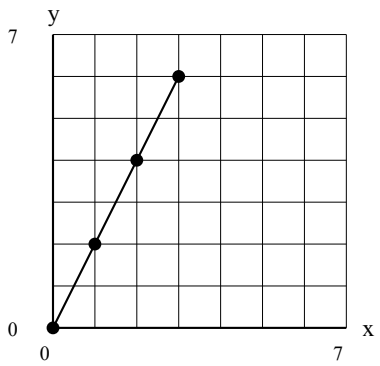
radius: Distance from the center to the circle. Example: A radius of 3 cm gives diameter 6 cm.

diameter: Distance across the circle through its center. Example: Diameter = 2 × radius.

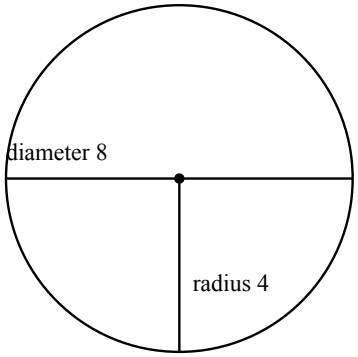
circumference: The distance around a circle. Example: $C = \pi d$

π: The ratio of a circle's circumference to its diameter. Example: π is about 3.14, not exactly 3.14.

How it works



The straight line y = 2x passes through the origin. A scaled copy multiplies every length by the same factor. Proportional relationships have the form y = kx.



A diameter contains two radii. Circumference measures the boundary. Circle area uses the square of the radius. π is exact; 3.14 is an approximation.

Worked examples

1. A copy of a 3 cm by 5 cm rectangle uses scale factor 2.

The new lengths are 6 cm and 10 cm.

The original area is 15 square cm; the copy area is 60 square cm.

The area factor is $60 \div 15 = 4 = 2^2$.

Answer: 6 by 10 cm; area 60 square cm

2. Find area when a circle’s diameter is 10 m.

The radius is half the diameter: 5 m.

Square the radius: $5^2 = 25$.

Area = 25π square m, or about 78.5 square m using 3.14.

Answer: 25π square m; about 78.5 square m

Common mistakes

Adds the same amount to every length. Why: Confuses additive change with scaling. Fix: Compare each copy-to-original ratio.

Uses the length factor for area. Why: Counts one changing direction. Fix: Build an array of scaled unit squares.

Calls any straight line proportional. Why: Misses the origin requirement. Fix: Find the output when the input is zero.

Optional preview

Signed multiplication follows equal-step patterns: $(-3) \times 4 = -12$. A quotient asks for the missing factor. Since $(-3) \times 4 = -12$, $(-12) \div (-3) = 4$.

Try it

1. A 3 by 4 cm rectangle is copied with scale factor 2. Find the new side lengths.

2. A 5 by 6 cm rectangle is copied with scale factor 3. Find the new side lengths.

3. A circle has diameter 3 cm. Find circumference exactly and using 3.14.

4. A circle has diameter 5 cm. Find circumference exactly and using 3.14.

5. A circle has diameter 7 cm. Find circumference exactly and using 3.14.

Answers

1. 6 cm and 8 cm 2. 15 cm and 18 cm 3. 3π cm; about 9.42 cm 4. 5π cm; about 15.70 cm 5. 7π cm; about 21.98 cm

Where this connects

The bridge joins ratios, equations, and geometry. Future lessons can revisit these ideas with more complex percent and signed-number problems.

Daily practice

A. Warm-up (3 minutes)

1. 12×3

Answer: _____

2. 12×7

Answer: _____

3. 12×4

Answer: _____

4. 12×9

Answer: _____

B. Practice

5. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

Answer: _____

6. 4 markers cost $2.60. Find dollars per marker.

Answer: _____

7. Find $10 - 1.286$.

Answer: _____

8. A prism has right-triangle ends with sides 3, 4, 5 cm and length 6 cm. Find surface area.

Answer: _____

Practice and thinking

Built-in practice: Scale, proportion and circles

9. Order −4, 0, −9 from least to greatest.

Answer: _____

10. Find the MAD of 1, 3, 5, 7.

Answer: _____

11. A 8 by 12 cm rectangle is copied with scale factor $\frac{1}{2}$. Find the new side lengths.

Answer: _____

12. A map uses 1 cm for 20 km. A route is 4 cm long. Find the actual distance.

Answer: _____

13. A shape has area 20 square cm. All lengths scale by $\frac{1}{2}$. Find the new area.

Answer: _____

14. A proportional table contains (6, 9). Find k in $y = kx$.

Answer: _____

C. Think

15. Eli triples every length of a drawing. He says its area also triples. Explain and repair his claim.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Write $\frac{1}{2}$ as a decimal and a percent.

Answer: _____

2. Write $\frac{1}{4}$ as a decimal and a percent.

Answer: _____

3. Write $\frac{3}{4}$ as a decimal and a percent.

Answer: _____

4. Write $\frac{1}{8}$ as a decimal and a percent.

Answer: _____

B. Practice

5. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

Answer: _____

6. Mix A has 2 cups juice and 3 cups water. Mix B has 3 cups juice and 5 cups water. Which has more juice for the same water?

Answer: _____

7. Find $3\ \frac{1}{2} \div \frac{3}{4}$.

Answer: _____

8. Solve $1.5x = 4.80$.

Answer: _____

Practice and thinking

Built-in practice: Scale, proportion and circles

9. Find |0| and describe it as a distance.

Answer: _____

10. Find the median of 8, 2, 4, 6.

Answer: _____

11. A circle has diameter 3 cm. Find circumference exactly and using 3.14.

Answer: _____

12. A circle has radius 2 cm. Find circumference exactly and using 3.14.

Answer: _____

13. A circle has radius 2 cm. Find area exactly and using 3.14.

Answer: _____

14. A circle has diameter 4 cm. Find area exactly and using 3.14.

Answer: _____

C. Think

15. Maya finds area from a diameter of 14 cm by writing $\pi \times 14^2 = 196\pi$. Explain and repair it.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. 12.2×10

Answer: _____

2. $1200 \div 100$

Answer: _____

3. Double 12.5

Answer: _____

4. Half of 72

Answer: _____

B. Practice

5. Compare $\frac{2}{4}$ and $\frac{2}{5}$.

Answer: _____

6. Yellow : purple is 2 : 5. There are 35 beads. Find both counts.

Answer: _____

7. Find 140% of 30 points.

Answer: _____

8. Which is the reciprocal of $2\ \frac{1}{2}$: $\frac{2}{5}$ or $\frac{5}{2}$? Explain using a product of 1.

Answer: _____

Practice and thinking

B. Practice, continued

9. Find 2.45 + 4.078.

Answer: _____

10. Solve 1.5x = 4.80.

Answer: _____

11. Find the surface area of a closed 1 by 4 by 5 cm box.

Answer: _____

12. Write an inequality for x that means less than 5. Is the boundary included?

Answer: _____

13. A: 3, 4, 5. B: 1, 4, 7. Compare their means and ranges.

Answer: _____

14. A circle has diameter 3 cm. Find circumference exactly and using 3.14.

Answer: _____

C. Think: weekly puzzle

15. A garden is 3 cm by 4 cm on a map where 1 cm means 10 m. Redraw it where 1 cm means 20 m. Find the new drawing dimensions and the actual garden area.

Answer: _____

Daily practice

A. Warm-up (3 minutes)

1. Round 1248 to the nearest hundred.

Answer: _____

2. Round 12.72 to the nearest whole number.

Answer: _____

3. Use 20 × 5 to estimate 19.8 × 5.1.

Answer: _____

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

Answer: _____

B. Practice

5. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

Answer: _____

6. Find 140% of 30 points.

Answer: _____

7. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

Answer: _____

8. Find 9.6 ÷ 8.

Answer: _____

Practice and thinking

Built-in practice: Data and statistics

9. Solve x + 14 = 31.

Answer: _____

10. A rectangle is 7 cm by 3 cm. Find its area.

Answer: _____

11. Find the mean of 1, 4, 5, 6.

Answer: _____

12. Find the median of 10, 3, 5, 7, 2, 9.

Answer: _____

13. Find the range of 0, 2, 5, 10.

Answer: _____

14. Find the MAD of 2, 5, 5, 8.

Answer: _____

C. Think

15. A map uses 1 cm for 50 km. A route is 2 cm long. Find the actual distance.

Answer: _____

16. Maya finds area from a diameter of 14 cm by writing $\pi \times 14^2 = 196\pi$. Explain and repair it.

Answer: _____

Stretch (optional)

17. Compute $(-3) \times 4$ and $(-12) \div (-3)$.

Answer: _____

18. A price rises from $20 to $25. Find the percent increase.

Answer: _____

Weekly check

20 minutes. No calculator. Show your work.

1. Find $\frac{3}{4} \times \frac{4}{5}$.

Answer: _____

2. Plot (1, 2), (2, 4), and (4, 8). Do they lie on a line through the origin?

Answer: _____

3. Find 25% of 32 stickers.

Answer: _____

4. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

Answer: _____

5. Find 0.25×0.04.

Answer: _____

Weekly check

6. Expand $3(x + 6)$.

Answer: _____

7. Find the surface area of a closed 1 by 4 by 5 cm box.

Answer: _____

8. Find the opposite of -1.

Answer: _____

9. Find the MAD of 3, 3, 9, 9.

Answer: _____

10. Eli triples every length of a drawing. He says its area also triples. Explain and repair his claim.

Answer: _____